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Technoökonomische Optimierung von CO₂ -Elektrolyse- und Separationssystemen zur
Syngasherstellung

Techno-Economic Optimization of CO₂ Electrolysis and Separation Systems for Syngas
Production

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Abstract

Current syngas production relies almost exclusively on coal and natural gas feedstocks. CO₂ electrolysis offers a promising low-carbon alternative, particularly through its direct use of CO₂ as a feedstock and its potential integration with renewable energy sources.

A large body of research has focused on optimizing the CO₂ to CO electrolysis unit by varying inlet concentration, power supply, and electrolyzer architecture. Comparatively little attention, however, has been paid to the consequences for the downstream separation processes.

This work combines CO₂ and H₂ electrolysis models with three alternative separation systems: pressure-swing adsorption (PSA), membrane separation, and amine washing. A techno-economic optimization provides deeper insight into the electrolysis and separation coupling and enables a cost comparison of the alternative flowsheets.

The work comprises the following work packages:

- Identification and selection of a suitable CO₂ electrolysis model
- Identification and selection of membrane separation and amine washing models
- Integration of the models into a combined flowsheet
- Optimization of the integrated flowsheet
- Comparison of the flowsheets under varying operating conditions

The base implementation of the PSA model was provided by previous work carried out with the same supervisor; it was corrected and extended as part of this thesis. The results are evaluated and discussed with respect to the cost-optimal separation route and the coupling between electrolyzer efficiency and downstream separation.

Aufgabenstellung

Die aktuelle Synthesegasproduktion stützt sich nahezu ausschließlich auf Kohle- und Erdgas-Einsatzstoffe. Die CO_2 Elektrolyse stellt eine vielversprechende kohlenstoffarme Alternative dar, insbesondere durch die direkte Nutzung von CO_2 als Einsatzstoff und das Potenzial zur Integration erneuerbarer Energiequellen.

Ein Großteil der Forschung hat sich auf die Optimierung der CO_2 -zu- CO -Elektrolyseeinheit konzentriert, indem Eingangskonzentration, Energieversorgung und Elektrolyseur-Architektur variiert wurden. Vergleichsweise wenig Aufmerksamkeit wurde jedoch den Auswirkungen auf die nachgelagerten Trennprozesse gewidmet.

Diese Arbeit kombiniert CO_2 - und H_2 -Elektrolysemodelle mit drei alternativen Trennsystemen: Druckwechseladsorption (PSA), Membrantrennung und Aminwäsche. Eine techno-ökonomische Optimierung liefert tiefere Einblicke in die Kopplung von Elektrolyse und Trennung und ermöglicht einen Kostenvergleich der alternativen Fließbilder.

Die Arbeit umfasst die folgenden Arbeitspakete:

- Identifikation und Auswahl eines geeigneten CO_2 -Elektrolysemodells
- Identifikation und Auswahl von Modellen für Membrantrennung und Aminwäsche
- Integration der Modelle in ein kombiniertes Fließbild
- Optimierung des integrierten Fließbilds
- Vergleich der Fließbilder unter verschiedenen Betriebsbedingungen

Die Basisimplementierung des PSA-Modells stammt aus einer vorherigen Arbeit, die mit demselben Betreuer durchgeführt wurde; sie wurde im Rahmen dieser Arbeit korrigiert und erweitert. Die Ergebnisse werden im Hinblick auf die kostenoptimale Trennrouten und die Kopplung zwischen Elektrolyseur-Effizienz und nachgelagerter Trennung ausgewertet und diskutiert.

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Chapters 3, 4, 5	Gemini, Claude Code	Generating Code to plot Results
Chapter 3	Claude Code	Assisting in wiring Flowsheets together
Conclusion Intro	Claude	Reformulating/ restructuring Text
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1 Introduction

Syngas ($\text{CO} + \text{H}_2$) is a key platform feedstock in today's process industry. Prominent syngas products include methanol, hydrogen and Fischer-Tropsch fuels [1]. Currently it is almost exclusively produced from coal and natural gas [2]. As the process industry moves to decarbonize, electrochemical processes powered by renewable electricity are a promising alternative [3].

Producing syngas electrochemically means generating both components from electricity. The hydrogen is supplied by water electrolysis, the carbon monoxide is produced electrochemically from CO_2 , using CO_2 from point sources or direct air capture and thereby closing the carbon loop [4].

Research on CO_2 electrolysis has mostly focused on optimizing the electrolyzer itself, varying inlet concentration, cell voltage and electrode or membrane architecture, while the downstream separation and the recycle of unreacted CO_2 have received far less attention. When techno-economic studies do include the separation, it is usually reduced to a composition-independent black box, whose cost scales with the total gas flow rate alone, even though Jouny et al. note that a more detailed separation model would shift the cost-optimal CO_2 conversion [5]. The separation step itself has only recently become a focus of direct study. Alerte et al. only look at the energy use of a single separation train for the CO_2 -to-ethylene case [6], while Lee et al. compare a membrane against an amine wash, but only for the anode-side CO_2 / O_2 stream and without pressure-swing adsorption [7]. No single study, however, combines all configurations. A consistent techno-economic comparison of the three main separation routes, pressure-swing adsorption, membrane separation and amine washing, for the cathode-side CO / CO_2 / H_2 syngas stream is still missing, in particular one that closes the recycle loop so the separation choice feeds back into the electrolyzer's cost-optimal operating point.

To close this gap, this thesis compares the three separation routes for electrochemically produced syngas on a consistent techno-economic basis. The comparison answers two questions: which route produces syngas at the lowest lifecycle cost, and how does the choice of separation feed back onto the electrolyzer's operating cost through the recycle? Answering these questions identifies both the cheapest separation route and the dominant cost driver of the integrated process. To this end, three flowsheets, one per separation route, are optimized for minimum lifecycle cost given a common syngas target.

This thesis is structured as follows: Chapter 2 reviews CO₂ electrolysis, the three separation technologies, and the existing techno-economic models and summarizes current relevant techno-economic research. Chapter 3 develops the integrated flowsheet and its submodels, including the CO₂ electrolyzer, the H₂ electrolyzer, the pressure-swing adsorption, membrane, and amine washing models. Chapter 4 examines the behavior of the CO₂ electrolysis surrogate and the pressure-swing adsorption model through sensitivity analyses. Chapter 5 presents and compares the cost-optimal operating points of the three flowsheets and varies the key economic parameters. Chapter 6 summarizes the findings and gives an outlook.

2 State of the Art

This chapter briefly reviews the current techno-economic landscape of literature about CO₂ electrolysis, explains CO₂ electrolysis types and models. Finally the three main contenders for downstream processing are explored.

2.1 CO₂ Electrolysis

This section focuses on CO₂ electrolysis. First possible electrolysis setups and membranes are explored in Section 2.1.1. Then the inner workings of a CO₂ electrochemical reactor are explored in Section 2.1.2 and key performance metrics are introduced in Section 2.1.3. Finally current standalone gas-diffusion electrolysis models are explored in Section 2.1.5 as well as their techno-economic counterparts in Section 2.1.4.

2.1.1 CO₂ Electrolysis Setups

Electrochemical reactors always consist of an anode and cathode halfcell. Both are linked through their respective anode and cathode electrodes which experience a difference in voltage to drive the electrochemical reactions [4]. The cathode side is fed CO₂, which is electrochemically reduced to the carbon-containing product. The anode side typically gets fed water (although other anolytes are being studied), which is oxidized and supplies the electrons and ions needed for the cathode reduction [8].

Depending on the catalyst material used different products can be targeted: Carbon Monoxide, Formic Acid, Methane, Ethylene, Ethanol, among others [5].

CO₂ electrochemical reactors vary substantially by general setup (H-cell, Zero-Gap,

Flow through) and membrane used (Anion Exchange Membrane (AEM), Cation (or Proton) Exchange Membrane (CEM, PEM), Bipolar Exchange Membrane (BPM)) [8, 4]. Figure 2.1 schematically shows the three setups.

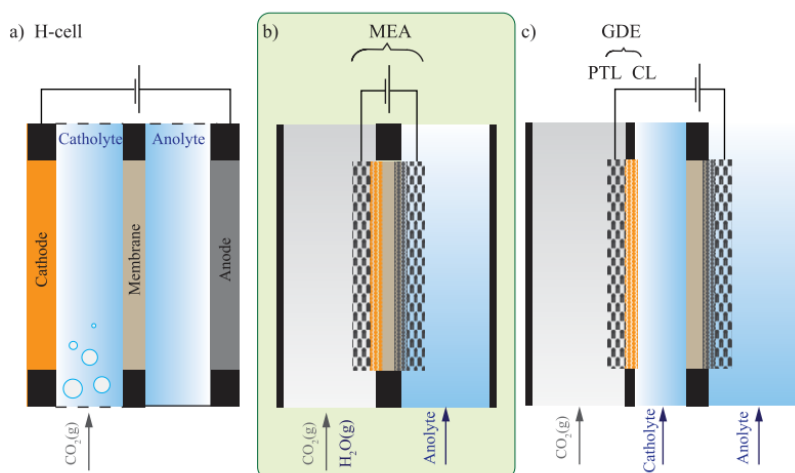


Figure 2.1: Sketch of a gas-fed batch cell/H-cell a), zero-gap flow-through cell with a membrane electrode assembly fed with a gaseous CO_2 water-vapor mix b) and flow-through cell fed with a liquid catholyte and gaseous CO_2 which are separated by a gas diffusion electrode (GDE) comprised of the porous transport layer (PTL) and the catalyst layer (CL) c) [8]

H-cell reactors are batch reactors while the other two configurations work continuously i.e. as continuous flow-cells. Membrane electrode assembly/Zero-Gap configurations are the main focus of research, ridding themselves of the catholyte required in gas-diffusion electrode configurations [8].

2.1.2 CO_2 Gas-diffusion Electrolysis with Anion Exchange Membrane

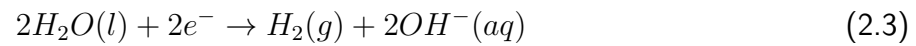
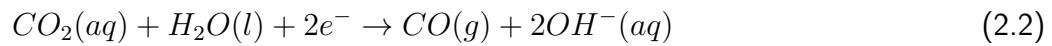
Depending on the combination of setups chosen the specific reactions vary slightly. This section specifically describes the inner workings of a gas diffusion electrode with an anion exchange membrane as this is the setup used in this thesis.

On the anode side the water gets oxidized to oxygen, electrons and hydrogen protons.



On the cathode side, a gaseous stream of mostly CO_2 passes through the channel. It meets a gas diffusion layer on the side wall of the channel which facilitates the dissolution of CO_2 into the catholyte.

At the catalyst layer, the CO_2 dissolved in the catholyte reacts with water and electrons to form CO . The catalyst material governs the selectivity: silver and gold, for instance, favor CO formation. A competing side reaction is the formation of hydrogen [4].



The dissolved potassium bicarbonate carries the ionic current and buffers the catholyte. At the same time, the alkaline environment at the cathode is unavoidable: the CO_2 reduction produces hydroxide ions, which react with the excess CO_2 to form bicarbonate and carbonate [4]:



HCO_3^- and CO_3^{2-} can pass through the membrane to the anode side, lowering conversion efficiency [9].

2.1.3 Key Performance Metrics

A few key metrics are used to assess the performance of a given electrolysis unit [4].

The total conversion refers to the fraction of CO_2 consumed [10]:

$$\chi_{tot} = \frac{c_{\text{CO}_2, in} - c_{\text{CO}_2, out}}{c_{\text{CO}_2, in}} \quad (2.6)$$

The heterogeneous conversion refers to the CO_2 converted to product, in the case of CO this resolves to [10]:

$$\chi_{het} = \frac{c_{\text{CO}, out}}{c_{\text{CO}_2, in}} \quad (2.7)$$

The homogeneous consumption is the remaining used CO_2 wasted to the anode side:

$$\chi_{hom} = \chi_{tot} - \chi_{het} \quad (2.8)$$

The faradaic efficiency refers to the fraction of the free electrons used for a given product and can be thought of as the analogous version to selectivity in traditional chemical engineering.

$$FE_i = \frac{\dot{n}_i z_i F}{I} \quad (2.9)$$

where $\dot{n}_i$ is the molar flow rate of product i , z_i is the number of electrons required to produce product i (2 for CO & H_2), F is Faraday's constant, and I is the total current.

The current density i describes the size of the electrolyzer vs the current applied:

$$i = \frac{I}{A} \quad (2.10)$$

where I is the total current and A is the electrode area.

λ_{CO_2} describes the stoichiometric ratio of CO_2 . It measures how much CO_2 is supplied vs stoichiometrically required by the electrolysis unit:

$$\lambda_{CO_2} = \frac{\dot{n}_{CO_2, \text{in}}}{I/(2F)} \quad (2.11)$$

with the mole flow of CO_2 fed into the unit $\dot{n}_{CO_2, \text{in}}$, total current I and faraday constant F .

2.1.4 Techno-Economic Assessment of CO_2 Electrolysis

Most techno-economic assessments (TEA) of CO_2 electrolysis operate at a high level of abstraction [11]. Faradaic efficiencies, current density, cell voltage and single-pass carbon efficiency are taken from lab measurements and passed through a simple black-box model to compute capital and operating expenditures [12]. The downstream separation is typically black box: its cost is commonly tied to the total gas flow rate alone, independent of the stream composition [5]. This is a strong simplification, since the separation cost depends on how much unreacted CO_2 the stream contains and on the technology used to remove it.

A notable step towards more detail is the work of Bagemihl et al. [10], who model the electrolysis unit itself in greater depth. They provide three electrolyzer models of increasing modeling depth (M1, M2, M3) and, combined with a simple cost correlation for a pressure-swing adsorption (PSA) unit, determine the cost and optimal operating conditions (electrolyzer size, cell voltage, power consumption) for ethylene production. The added depth, however, sits entirely on the electrolyzer side; the separation remains a blackbox cost correlation.

Only recently has the downstream separation itself received closer attention. Alerte et al. [6] model a mature separation train (amine CO_2 removal, glycol dehydration and cryogenic distillation) for the CO_2 to ethylene case and quantify its energy intensity. They show that the cell performance parameters strongly govern the downstream separation cost, but they do not optimize the system economically nor compare separation technologies, and they note that PSA is poorly suited to their ethylene tail gas. Lee et al. [7] compare a multi-stage membrane unit against a piperazine amine wash for recovering the CO_2 that crosses over to the anode, and find the two economically comparable. Their comparison, however, covers only the anode-side CO_2 / O_2 stream, treats the separation as a standalone unit with a fixed feed, and does not include PSA. Additionally, neither study examines the cathode-side CO / CO_2 / H_2 syngas stream, nor couples the separation back to the electrolyzer operating point through the recycle.

Across these studies the separation is therefore either reduced to a composition-independent cost correlation, assessed for a single technology and product, or analyzed in isolation from the electrolyzer it serves. What is missing is a consistent techno-economic optimization that compares the three principal separation routes, PSA, membrane separation and amine washing, for the cathode-side CO / CO_2 / H_2 syngas stream, and closes the recycle loop so the separation choice feeds back into the electrolysis unit.

2.1.5 CO₂ Gas-diffusion Electrolysis Models

Weng et al. [13] developed a multiphysics model of the gas-diffusion electrode cell.

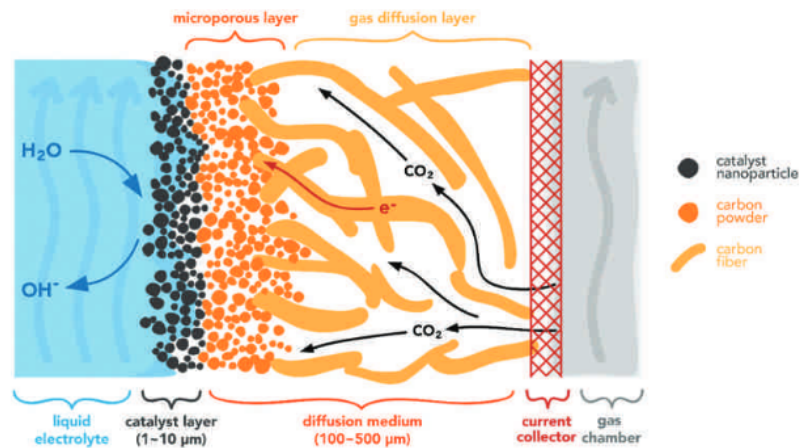


Figure 2.2: Cathode Side of CO₂ electrolysis unit by Weng et al. [13]

A gas diffusion electrode generally consists of a catalyst layer (CL), where the reaction takes place, and a porous diffusion medium (DM) that transports the gaseous CO₂ to it. Weng et al. describe both layers in a one-dimensional model as shown in figure 2.2. The bulk liquid electrolyte is modeled as a boundary condition with fixed bulk flow rate and concentrations. The gas chamber on the other side is also a boundary composition with fixed CO₂ composition. They released all the equations for their model but no code.

Bree et al. [14] developed a modular dynamic model framework in gProms which allows for different electrolyzer setups.

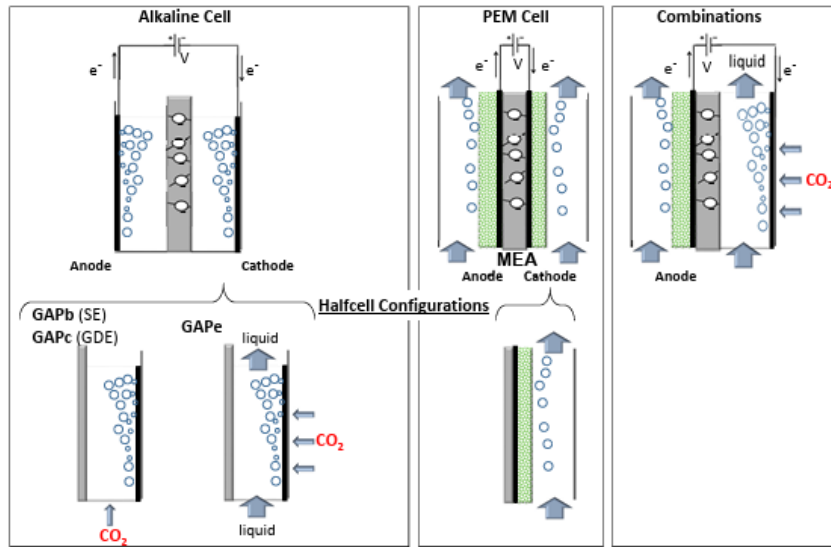


Figure 2.3: Scheme of different reactor setups [14]

Their 1D model framework allows for different gas-diffusion electrode setups or even zero-gap configurations (PEM). In contrast to Weng et al. [13], they model the full cell but do not spatially resolve the catalyst layer in detail.

As previously mentioned, Bagemihl et al. [10] created three models of the electrolysis unit with varying modeling depths. The most detailed model, they named the M3 model is shown in the figure below

Full channel model M3

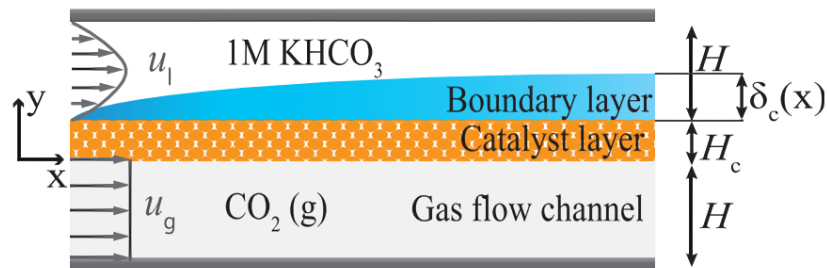


Figure 2.4: Full Channel Model (M3) from Bagemihl et al. [10]

The M3 model is a 1D Channel model composed of submodels for a gas flow channel, catalyst and boundary layer. The model is formulated as a boundary value problem, implemented in Matlab. A method of lines technique solves the concentration profile

of CO_2 , CO , and H_2 in the gas phase for given inlet concentrations and process conditions.

2.2 CO₂ Separation

2.2.1 Pressure Swing Adsorption

In Pressure Swing Adsorption (PSA) a multicomponent feed gas is pressurized and fed into a column packed with a solid adsorbent. The targeted species (CO₂) adsorbs at a faster rate and higher loading onto the solid, separating it from the remaining gas. Once the column approaches saturation, the pressure is lowered to atmospheric or sub-atmospheric levels (Vacuum Swing Adsorption, VSA), shifting the adsorption equilibrium and desorbing the CO₂ back into the gas phase [15].

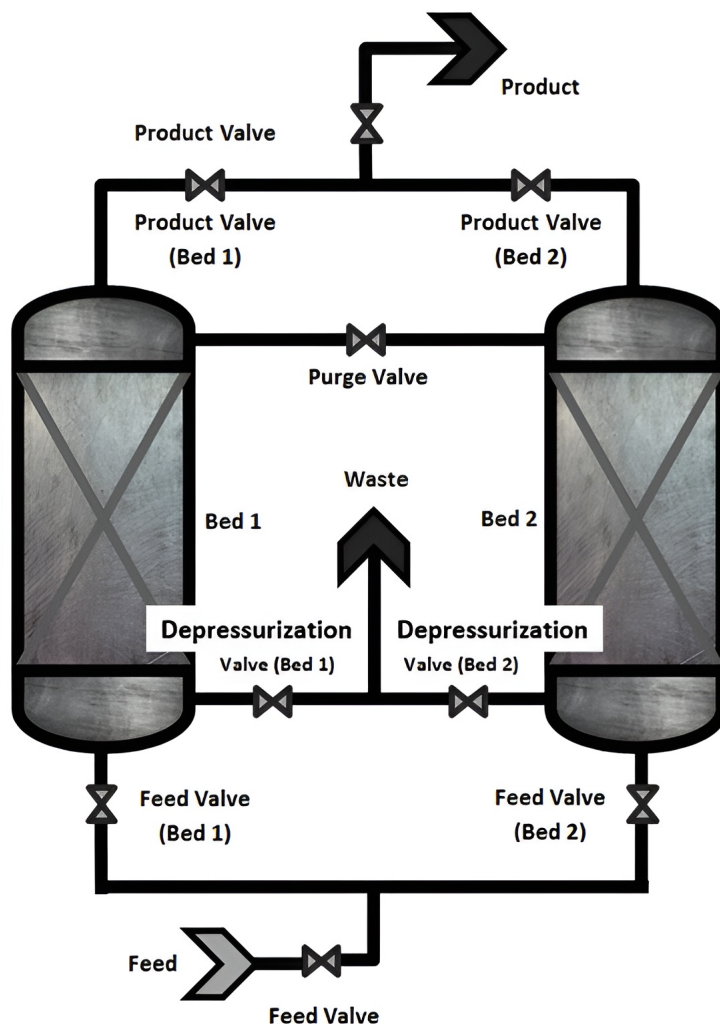


Figure 2.5: Two-bed PSA flowsheet Figure by Lopez et al.[15]

Figure 2.5 shows a typical two-bed setup based on the Skarstrom cycle: while one bed adsorbs at high pressure, the second regenerates at low pressure, and the beds alternate roles periodically. Operating at least two columns in parallel ensures a quasi-continuous product stream. After an initial transient, the system reaches a cyclical steady state (CSS), where the concentration and loading profiles within each column are identical from one cycle to the next. This CSS assumption allows a time-averaged steady-state model to replace a full dynamic simulation, greatly reducing computational cost.

In the context of CO₂ electrolysis, the PSA feed contains CO₂, CO, and H₂. Zeolite 13X is commonly selected as the adsorbent because it shows a high selectivity for CO₂ over CO and H₂, making it well suited for recovering unreacted CO₂ from the electrolyzer outlet [16].

Adsorption Equilibrium: Sips Isotherm

The equilibrium loading q_i^* of species i on the adsorbent is described by the Sips (Langmuir–Freundlich) isotherm. For a pure component, of species i , it reads:

$$q_i^* = q_{m,i} \frac{(b_i p_i)^{n_i}}{1 + (b_i p_i)^{n_i}} \quad (2.12)$$

where $q_{m,i}$ is the saturation capacity, b_i is the affinity constant, p_i the partial pressure of species i , and n_i the heterogeneity exponent. For a gas mixture it can be extended to the multicomponent Sips isotherm:

$$q_i^* = q_{m,i} \frac{(b_i p_i)^{n_i}}{1 + \sum_j (b_j p_j)^{n_j}} \quad (2.13)$$

The temperature dependence of b_i and $q_{m,i}$ are expressed through Arrhenius-type equations. The isotherm parameters for CO₂ and CO on zeolite 13X are taken from the literature [16].

Virtual Moving Bed Model

Sees et al. [16] propose a virtual moving bed (VMB) approach to model PSA at cyclical steady state (CSS). Rather than computing concentration gradients through the packed beds, column is treated as a steady-state component splitter: the feed is divided into a CO enriched product stream and a CO₂ enriched retentate stream based on adsorption equilibrium and efficiency parameter, which accounts for non ideal equilibrium loading caused by mass transfer resistance.

For a standard Skarstrom cycle, two such splitters are connected in series as shown in figure 2.6: the left splitter represents the high-pressure adsorption step, the right represents the low-pressure desorption step, and the virtual moving bed is formed by virtual internal bed hold-up streams, representing the material retained in the bed between operating states, that cycle between them.

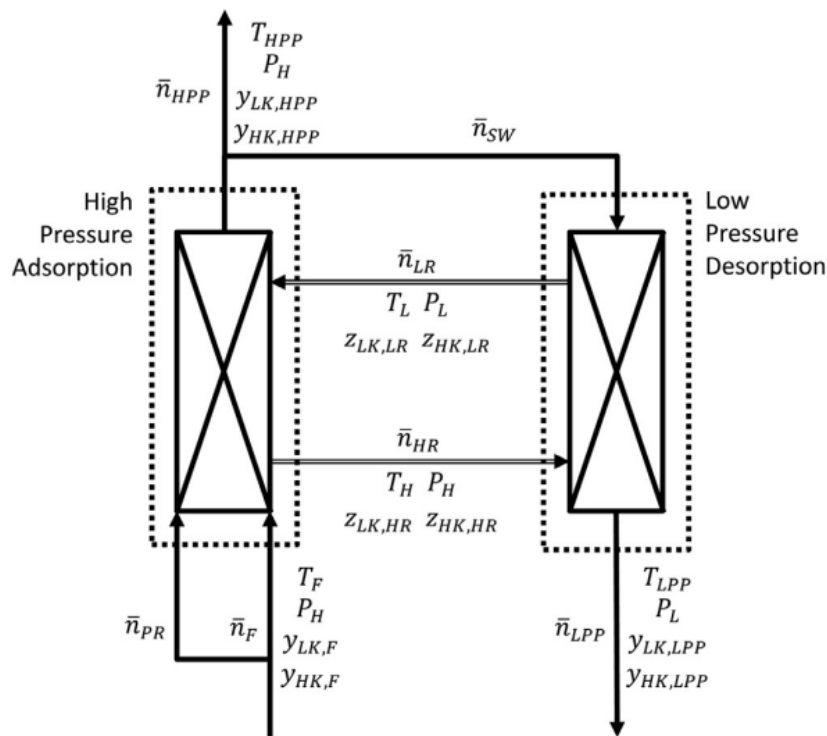


Figure 2.6: Virtual moving bed flowsheet for a two-bed Skarstrom PSA cycle. The left and right blocks represent the high-pressure adsorption and low-pressure desorption operating states respectively; double arrows denote adsorbate-containing retentate streams [16].

2.2.2 Amine Washing

Basic Flowsheet

Amine washing or amine scrubbing is a mature and well studied process for the separation of CO_2 and hydrogen sulfide from a gas stream. Its main application has been in flue gas treatment; the removal of CO_2 and hydrogen sulfide from coal and natural gas power plants [17].

A liquid mixture of water and amine, called the lean mixture, is fed into a washing column with the gas. Inside the washing column, CO_2 is absorbed from the gas phase into the liquid phase through an exothermal reaction with the amine. The resulting liquid, loaded with CO_2 , is called rich mixture and regenerated at higher pressure in a rectification column [18].

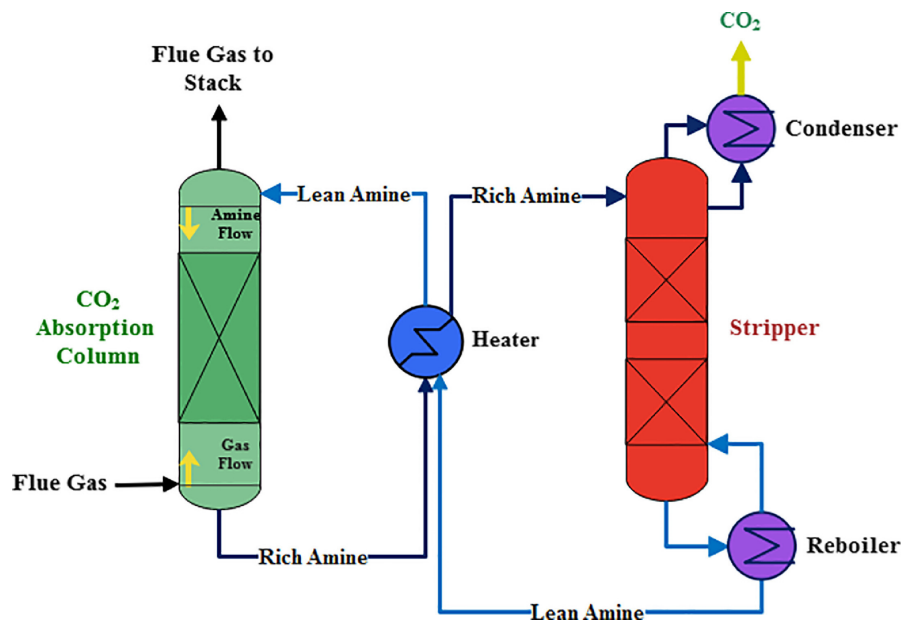


Figure 2.7: Simplified Amine Washing Flowsheet [19]

The process economics are dominated by the reboilers energy consumption which can account for up to half of the annualized process costs [19].

As a consequence much of the research has been focused on heat recovery and formulation of new amines or amine mixtures with better loading properties.

Common Amines & Absorption Reaction

Researchers have identified a lot of amines over the years, each with their niche advantages and disadvantages with regards to regeneration cost, working capacity, stability, absorption rate, volatility, and resistance to degradation. Nevertheless, many remain commercially unused and sourcing them would possibly require a separate production plant since amines are prone to degradation. We list the most commonly used amines, and piperazine (PZ), in the table below.

Table 2.1: Common amines for CO₂ capture [19, 20]

Amine	Industrial Maturity	Advantages
Monoethanolamine (MEA)	Old industry standard	Low cost, widely available
Diethanolamine (DEA)	Common replacement for MEA	Lower regeneration cost, better reactivity
N-Methyldiethanolamine (MDEA)	New industry standard	Lower regeneration cost than DEA, less degradation
Piperazine (PZ)	Novel	High loading, low regeneration cost, fast reaction

BASF is the first big commercial adopter of aMDEA which is a blend of MDEA and PZ. The University of Texas Separation Research Program has been experimenting with pure PZ in their pilot plant since 2015 [21].

2.2.3 Gas Membrane Separation

Gas permeation membranes separate a feed stream by exploiting differences in permeability through a dense polymer film. A species dissolves into the membrane on the high pressure feed side, diffuses through the membrane and desorbs on the low pressure permeate side [22]. The remaining feed stream that did not pass through the membrane is called the retentate. Figure 2.8 shows the setup.

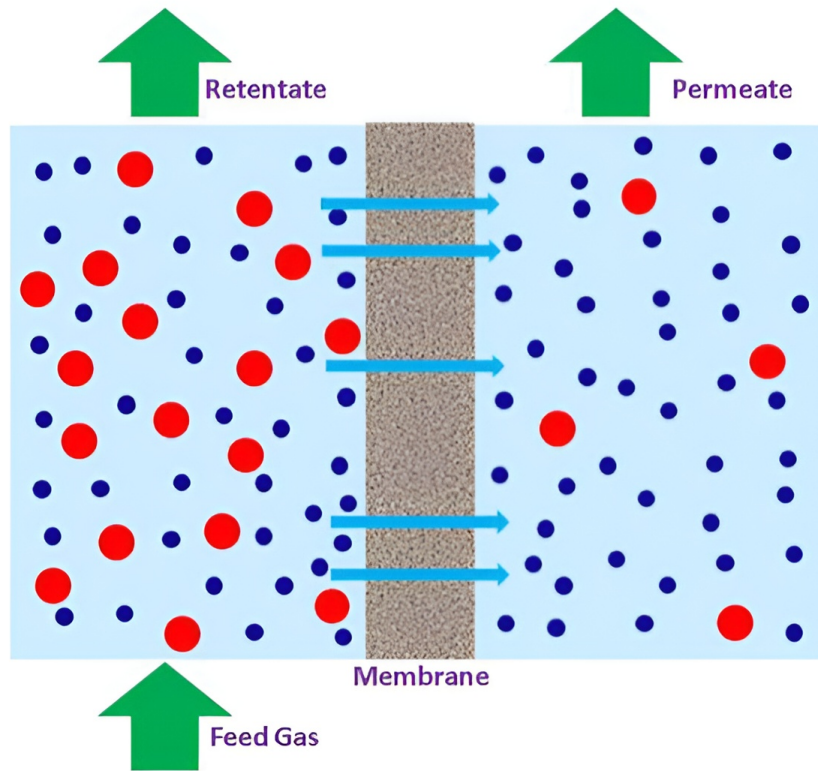


Figure 2.8: Basic Membrane Diagram [22]

Hesselmann et al. modeled the species flux through the membrane using the solution-diffusion model [23]:

$$J_i = Q_i \cdot \delta \cdot (y_{r,i} p_r - y_{p,i} p_p) \quad (2.14)$$

where Q_i is the permeance of species i , $\delta = 0.5 \mu\text{m}$ is the thickness of the selective layer, $y_{r,i}$ and $y_{p,i}$ are the molar fractions in the retentate and permeate, and p_r and p_p are the respective pressures.

Hesselmann et al. list the following membrane permeabilities in their supporting information:

Table 2.2: Permeabilities of membrane materials used in the downstream separation [23]

Material	P_{CO_2}	P_{CO}	P_{H_2}	P_{O_2}
	[barrer]			
Polyimide	13	1	50	–
PEBAX	55.85	1.39	32.11	4.69
Matrimid	0.25	3.46	13.53	–
PRISM	0.25	5.6	14	–

With $Q_i = P_i/\delta$ relating permeabilities to the transport equation (2.14).

Hesselmann et al. assume a fixed capital cost per m^2 for the membrane :

$$C_{\text{CAPEX, mem}} = BC_{\text{membrane}} A_{\text{mem}} \quad (2.15)$$

where $BC_{\text{membrane}} = 50/\text{m}^2$ is the area-specific base cost for polymeric membranes and A_{mem} is the required membrane area from the optimization.

For OPEX, membrane replacement is included at a assumed lifetime of 5 years, depreciated over the 20-year plant lifetime.

$$\dot{C}_{\text{OPEX, mem}} = d_{\text{year}} \cdot t_{\text{plant}} \cdot C_{\text{CAPEX, mem}} \left(\frac{1}{t_{\text{lifetime, membrane}}} + f_{\text{maint}} \right) \quad (2.16)$$

where $f_{\text{maint}} = 2.5\%$ is the annual maintenance cost as a fraction of $C_{\text{CAPEX, mem}}$, d_{year} is the annual depreciation factor, $t_{\text{plant}} = 20 \text{ a}$ is the plant lifetime and $t_{\text{lifetime, membrane}} = 5 \text{ a}$ is the assumed membrane lifetime.

3 Model Development

3.1 General Flowsheet Setup

CO₂ electrolyzers are known to run more efficiently with an excess of CO₂ at the inlet. The general structure of the flowsheet is the same for all three separation techniques (Membrane, PSA, Amine Washing), with only the separation block differing on the used separation technique.

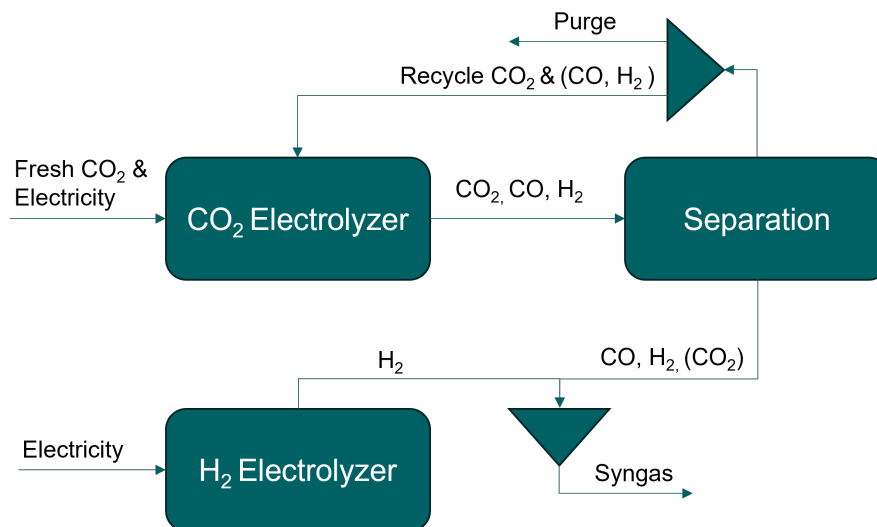


Figure 3.1: General Flowsheet Setup

The purge is kept constant at 0.02 to increase solver convergence and ensure comparability across flowsheets. The syngas target is set to a H₂ to CO ratio of 2:1 and targets 75 t/day.

The solver looks for a converged operating point x_{opt} under the constraint of minimizing the lifecycle cost of the entire flowsheet:

$$\min_{\mathbf{x}_{\text{opt}}} \left\{ C_{\text{lifecycle}} = C_{\text{CAPEX}} + \text{PVF} \dot{C}_{\text{OPEX}} \right\} \quad (3.1)$$

C_{CAPEX} includes the capital expenditure for the CO_2 electrolyzer, H_2 electrolyzer and the separation unit. $\dot{C}_{\text{OPEX}}$ represents the yearly operational expenditure of CO_2 and H_2 electrolyzer (electricity consumption) and the separation unit (electricity/steam consumption) and the cost of fresh CO_2 . Both split into the contributions of the separation unit, the H_2 electrolyzer and the CO_2 electrolyzer, with the fresh CO_2 feed entering the operational expenditure at the flowsheet level:

$$C_{\text{CAPEX}} = C_{\text{CAPEX, sep}} + C_{\text{CAPEX, H}_2, \text{elec}} + C_{\text{CAPEX, CO}_2, \text{elec}} \quad (3.2)$$

$$\dot{C}_{\text{OPEX}} = \dot{C}_{\text{OPEX, sep}} + \dot{C}_{\text{OPEX, H}_2, \text{elec}} + \dot{C}_{\text{OPEX, CO}_2, \text{elec}} + \dot{C}_{\text{CO}_2, \text{feed}} \quad (3.3)$$

The present value factor (PVF) accounts for inflation:

$$\text{PVF} = \frac{1 - (1 + r)^{-n}}{r} \quad (3.4)$$

where $r = 0.085$ is the discount rate and $n = 20$ is the plant lifetime in years.

The design variables are categorized into flowsheet level and unit level variables:

$$\mathbf{x}_{\text{opt}}^T = \left[x_{\text{flowsheet}} \quad x_{\text{CO}_2, \text{elec}} \quad x_{\text{H}_2, \text{elec}} \quad x_{\text{sep}} \right] \quad (3.5)$$

3.2 Surrogate modeling of CO_2 Channel Electrolysis

Compute Time Black Box Approach

The electrolysis model by Bagemihl et al. [10] is chosen for its relative simplicity, but relevant techno-economic scope. In the Bagemihl paper the M3 full channel model is studied for ethylene electrolysis. Bagemihl et al. note that their framework "can be applied to various electrolyser designs and CO₂ -reduction products, such as CO or formate." [10] and provide a fully specified model for the CO case.

Full channel model M3

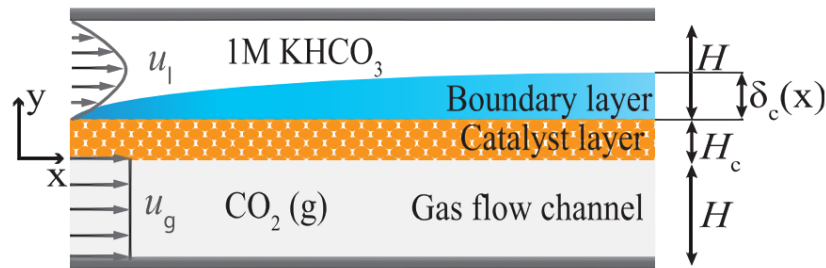


Figure 3.2: Full Channel Model (M3) from Bagemihl et al.[10]

The M3 model is a 1D Channel model composed of submodels for a gas flow channel, catalyst and boundary layer. The model is formulated as a boundary value problem, implemented in Matlab. A method of lines technique solves the concentration profile of CO₂, CO and H₂ in the gas phase for given inlet concentrations and process conditions.

Since this thesis implements the overarching syngas production flowsheets in Python using Pyomo, simply translating the M3 model to Python does not work as the iterative method of lines solver clashes with the Pyomo framework.

A second option would be a blackbox approach, where the flowsheet solver is only able to request computation of individual data points and its derivatives. After closer inspection this was found to be computationally intractable: On average computing 1 operating point requires around 40 seconds of execution time. From [24] it is known that during the optimization of the electrolyzer + PSA flowsheet the solver iterates ~450 times. With multiple initializations of the flowsheet (multistart) this would roughly take:

$$\begin{aligned}
T_{\text{total}} &= N_{\text{multistart}} \times N_{\text{solver}} \times T_{\text{MATLAB}} \\
&= 3 \times 450 \times 40 \text{ s} \approx 55,000 \text{ s} \approx 15.3 \text{ h}
\end{aligned} \tag{3.6}$$

This does not even assume the need for derivatives of the electrolysis model with respect to its inputs, which would again at least quadruple the computational cost.

Thus, a surrogate model is generated. Table 3.1 and table 3.2 below show the varied inputs and fixed parameters respectively.

Variables & Parameters

Table 3.1: CO₂ Electrolysis Input Variables

Variable Name	Variable	Unit	Lower Bound	Upper Bound
CO ₂ Inlet Concentration Share	$CO2_{share}$	-	0.2	1
Cathode Voltage	E_{appl}	V	-1.4	- 0.5
Gas flow speed	v_{gas}	$\frac{ml}{min}$	5	50

$CO2_{share}$ are set deliberately wide to not constrain the optimizer during the flowsheet optimizations. Cathode Voltage and Gas flow speed bounds are set according to the range studied by Kas et al. [25], used by Bagemihl et al. as a stepping stone for their electrolysis model.

The electrolysis unit is considered to only be fed CO₂ and CO at the inlet since the M3 model cannot handle additional H₂ input. The H₂ mole fraction at the inlet is expected to be small anyway since H₂ output is considerably smaller than CO and gets largely separated out by the separation unit that follows. The small amount of H₂ fed back to the electrolyzer from the recycle bypasses the electrolyzer entirely and gets fed into the electrolyzer outlet. This conveniently allows for a reduction in the complexity of the surrogate modeling problem. According to Dalton's and ideal gas law the concentrations then resolve to:

$$c_{CO_2,in} = \frac{P}{RT} x_{CO_2} \quad (3.7)$$

$$c_{CO,in} = \frac{P}{RT} (1 - x_{CO_2}) \quad (3.8)$$

$$c_{H_2,in} = 0 \quad (3.9)$$

Table 3.2: CO₂ Electrolysis Fixed Parameters

Variable Name	Variable	Unit	Value
CO ₂ Channel Height	H	mm	1
Channel Width	W	mm	10
Channel Length	L	mm	100
Catalyst Layer Thickness	H_c	mm	3.6×10^{-3}
Inlet Pressure	P	bar	1
Electrolyte Flow Speed	u_l	$\frac{ml}{min}$	1

Increasing the pressure is not part of the scope of the M3 model and cannot be justifiably varied. Channel geometry, electrolyte flow speed and ohmic losses are kept constant as specified by Bagemihl.

Voltage - Current Relationship

Since the voltage-current relationship in the mechanistic model is already expressed in closed-form analytical equations, these are adopted directly into the surrogate model without modification:

$$V_{cell} = E_{anode} + \eta_{act,A} - E_{appl} + \eta_{ohm} \quad (3.10)$$

$$\eta_{act,A} = \frac{RT}{0.5F} \operatorname{arcsinh} \left(\frac{i}{2 \times 10^{-7}} \right) \quad (3.11)$$

$$\eta_{ohm} = i \left(\frac{H_{ch}}{\sigma_{el}} + \frac{L_m}{\sigma_m} \right) \quad (3.12)$$

$$i = BV_{i,CO} \frac{\bar{c}_{CO_2}}{BV_{CO_2,ref}} \exp \left(-\frac{BV_{CO} F}{RT} (E_{appl} + BV_{ECO}) \right) + BV_{i,H_2} \exp \left(-\frac{BV_{H_2} F}{RT} E_{appl} \right) \quad (3.13)$$

$$\bar{c}_{CO_2} = \frac{c_{CO_2,in} + c_{CO_2,out}}{2} k_H \quad (3.14)$$

$$I = i A_{total} \quad (3.15)$$

$$P = I V_{cell} \quad (3.16)$$

The cell voltage V_{cell} is composed of the anode equilibrium potential E_{anode} , the anodic activation overpotential $\eta_{act,A}$, the applied voltage E_{appl} , and the ohmic overpotential η_{ohm} . The cathode equilibrium potential BV_{ECO} enters through the current density rather than the cell voltage, to avoid double-counting the CO standard potential.

Ohmic losses are computed from the electrode thickness H_{ch} , electrode conductivity σ_{el} , membrane thickness L_m and membrane conductivity σ_m .

The current density i is modeled through the Tafel approximation of the Butler-Volmer kinetics.

The mean cathode CO_2 concentration $\bar{c}_{CO_2}$ is the arithmetic mean of inlet and outlet concentrations scaled by the Henry constant $k_H = 0.85$.

The total current I and power consumption P follow from the current density and the total electrode area A_{total} .

Data Generation

1000 training data points are generated, varying the 3 input variables listed in table 3.1

based on Latin Hypercube Sampling (LHS) and 1000 test data points by randomly sampling each point in the space.

The boundary value problem (BVP) is solved using Matlabs inbuilt Euler method. As a fallback alternative the Heun method is attempted if the former does not converge. The training and test sets have a convergence failure rate of 3.8% and 41.1% respectively.

Training

The goal is to find simple relationships between inlet concentrations and voltage versus power consumption and outlet concentrations. To this end, symbolic regression implemented through the PySR python package developed by Cranmer Miles [26] is used. Symbolic regression has been developed for multi input - single output functions, forcing each outlet concentration to be trained and predicted individually.

Table 3.3: Hyperparameters of the PySR symbolic regression search.

Hyperparameter	Value
Loss function	L_1 (L1DistLoss)
Iterations	4000
Maximum complexity	30
Binary operators	+, −, ×, /, ^
Unary operators	square, cube, sqrt, cbrt, exp, log
Power constraint	^ : (−1, 2)
Nested constraints	No Nesting
Parsimony	0.01
Model selection	best

An absolute-error loss (L_1) was used to limit the influence of outliers. The search ran for 4000 iterations; expression size was capped at 30 nodes and a parsimony coefficient of 0.01 penalised complexity, favouring compact and interpretable models. The operator set (arithmetic operations, exponentiation, and the unary operators square, cube, sqrt, cbrt, exp, log) reflects the polynomial, root and exponential behaviour expected from the underlying electrochemistry. To keep expressions well-behaved, the exponent complexity was restricted ($^ : (-1, 2)$) and nested constraints prevents unstable operator compositions (e.g. exp inside exp or log inside exp). The final model was selected with the best criterion, balancing accuracy against complexity along the Pareto front. For more details the reader is referred to the PySR paper [26].

The resulting surrogate equations have the following structure:

$$c_{CO_2,out} = \left((E_{appl} ((v_{gas}^{1/4} + E_{appl}) \theta_{1,1} - \theta_{1,2}))^3 + \theta_{1,3} \right) CO2_{share} \quad (3.17)$$

$$c_{CO,out} = CO2_{share} \left(\frac{\theta_{2,1}}{E_{appl} - v_{gas}^{\theta_{2,2}}} - (\theta_{2,3} E_{appl})^{\theta_{2,4}} + \theta_{2,5} E_{appl} \right)^{\theta_{2,6}} \quad (3.18)$$

$$c_{H_2,out} = \frac{(\theta_{3,1} E_{appl}^2 + \theta_{3,2})^{\theta_{3,3}}}{v_{gas}} \quad (3.19)$$

$\theta_{i,j}$ are constants estimated by the symbolic regression. The optimizer found $c_{CO_2,out}$, $c_{CO,out}$ to be functions of the applied voltage at the electrode E_{appl} , gas speed v_{gas} and share of gas feed containing CO_2 $CO2_{share}$. $c_{H_2,out}$ is found to have no dependence on the inlet gas composition.

3.3 Fully sized CO_2 Electrolysis Model

The voltage equations 3.10 - 3.16 of Bagemihl et al. [10] combined with the surrogate modeling equations 3.17 - 3.19 developed in section 3.2 formulate a model for a single channel.

A few sizing equations are needed to formulate a full electrolysis model containing multiple channels. The volumetric gas flow rate $\dot{V}$ through all N channels is:

$$\dot{V} = v_{gas} H W N \quad (3.20)$$

Molar flow rates are then obtained by:

$$\dot{n}_k = c_k \dot{V}, \quad k \in \{CO_2, CO, H_2\} \quad (3.21)$$

The total electrode area and current are:

$$A_{\text{tot}} = N W L, \quad I = i A_{\text{tot}}, \quad P_{\text{el}} = I V_{\text{cell}} \quad (3.22)$$

With reaction area of a single channel A_{cell} , total reaction area A_{tot} , current density i , current I , and power consumption P_{el} denoting the respective quantities.

Capital and yearly operating costs are formulated as:

$$C_{\text{CAPEX, CO}_2, \text{elec}} = c_{\text{area}} A_{\text{tot}} \quad (3.23)$$

$$\dot{C}_{\text{OPEX, CO}_2, \text{elec}} = P_{\text{el}} p_{\text{el}} + \frac{f_{\text{maint}} C_{\text{CAPEX, CO}_2, \text{elec}}}{h_{\text{op}}} \quad (3.24)$$

CAPEX are estimated by scaling the total electrolyzer area by area cost c_{area} and opex include maintenance and electricity cost. CO_2 input costs are included on the flowsheet level since some of it will be recycled CO_2 .

Table 3.4: Economic parameters of the CO_2 Electrolyzer Model

Parameter	Symbol	Value
Area Cost	c_{area}	4670 EUR/m ²
Electricity Price	p_{el}	0.08 EUR/kWh
Annual Operating Hours	h_{op}	8,400 h/a
Maintenance Factor	f_{maint}	0.03

With c_{area} found to be a reasonable estimate for large electrolysis plants by Steenbuck and von der Drecken [27].

3.4 H₂ Electrolysis Model

The H₂ electrolysis model is taken from Simone et al. [28]. It is a PEM (Polymer Electrolyte Membrane) water electrolyzer that boils down to a single fitted specific capital cost equation in EUR/kW:

$$C_{kW} = \kappa_3 \ln(P_{hp})^3 + \kappa_2 \ln(P_{hp})^2 + \kappa_1 \ln(P_{hp}) + \kappa_0 \quad (3.25)$$

where the fitted coefficients κ_0 – κ_3 are given in table A.3.

$$C_{\text{CAPEX, H}_2, \text{elec}} = C_{kW} P_{hp} f_{\text{stack}} \frac{\text{CEPCI}_{2024}}{\text{CEPCI}_{2021}} C_{F,el} \quad (3.26)$$

By multiplying the specific capital cost by the power draw, P_{hp} , the term gives the total installed CAPEX of the electrolyzer. The stack replacement factor $f_{\text{stack}} = 1 + (1 + r)^{-10}$ accounts for the discounted replacement of the PEM stack, expected around year 10 of operation, with r the discount rate. The cost-index ratio $\text{CEPCI}_{2024}/\text{CEPCI}_{2021}$ adjusts for inflation. Finally, $C_{F,el} = 1.75$ is the installation factor that scales the bare equipment cost to a fully installed cost, covering civil works and on-site installation.

Operating expenditure combines the annual electricity bill at price c_{el} over h_{op} operating hours with annual maintenance, taken as a fixed fraction f_{maint} of the installed capital:

$$\dot{C}_{\text{OPEX, H}_2, \text{elec}} = \underbrace{h_{op} c_{el} P_{hp}}_{\text{electricity}} + \underbrace{f_{\text{maint}} C_{\text{CAPEX, H}_2, \text{elec}}}_{\text{maintenance}} \quad (3.27)$$

The electrolyzer is characterized by a specific energy consumption e_{H_2} , obtained from the higher heating value of hydrogen $\text{HHV}_{\text{H}_2} = 39.4 \text{ kWh/kg}$ derated by the overall electrical efficiency $\eta_{el} = 0.64$:

$$e_{H_2} = \frac{HHV_{H_2}}{\eta_{el}} \approx 61.6 \text{ kWh/kg}_{H_2}. \quad (3.28)$$

Dividing the power draw by this specific consumption gives the hydrogen mass flow:

$$\dot{m}_{H_2} = \frac{P_{hp}}{e_{H_2}} \quad (3.29)$$

3.5 Corrections and Extension of Pressure Swing Adsorption

Sees et al.[16] propose a framework to build a pressure swing adsorption (PSA) model. Subsequently Tjahjana[24] used and implemented the framework to separate a CO₂ CO feedstream by combining it with isotherms measured by Park et al.[29].

Several errors were made in the implementation of the PSA model as envisioned by Tjahjana[24]. This Chapter denotes the major fixes made and describes the necessary extension to the kinetics.

3.5.1 Implementation Errors

Pressure Vessel Cost Correlation

The pressure vessel cost was estimated using the Guthrie approach [30]. As the source is not widely available the reader is referred to Douglas (1988) [31] appendix section E2, page 574 which sources from Guthrie.

Tjahjana [24] seems to have sourced its information from the 'Prozessentwicklung in der Verfahrenstechnik' lecture by Mitsos [32] which it in turn claims to source from Guthrie [30].

The original Guthrie correlation as presented by Douglas [31] gives the base purchasing cost of a pressure vessel as:

$$C_p = \frac{M\&S}{280} 101.9 D^{1.066} H^{0.802} \quad (3.30)$$

where D is the vessel diameter (ft), H is the vessel height (ft), $M\&S$ is the Marshall & Swift Equipment Cost Index (base value 280, circa 1968).

The cost correlation for pressure vessels given by Guthrie as claimed by Mitsos [32] is:

$$K_G = K_0 \left(\frac{L}{L_0} \right)^\alpha \left(\frac{D}{D_0} \right)^\beta \quad (3.31)$$

where L is the vessel length, D is the vessel diameter, and the parameters K_0 , L_0 , D_0 , α , β are given in table 3.5.

Pressure Vessel Cost Correlation

Table 3.5: Guthrie cost correlation parameters for pressure vessels [32]

Apparat	K_0 (10^3 \$)	L_0	D_0	α	β	MF2
Vertikaler Druckbehälter	1000	4.0	3.0	0.81	1.05	4.23
$1 < D < 10$ ft, $4 < L < 100$ ft						

Assuming a height of 100 ft and diameter of 10 ft, the Douglas cost correlation comes out to $C_p \approx 47,500\$$. For equation (3.31) the base vessel cost comes out to $K_G \approx 50400$. But crucially, table 3.5 lists K_0 to have a unit of 1000\$ implying K_G is also priced in 1000\$ and the actual vessel cost to be $\approx 50,400,000\$$.

This is the likely mistake Tjahjana made, inflating the vessel cost by a factor 10^3 . The mistake is corrected in the new implementation.

Vacuum Pump

The original model sized the vacuum pump as a standard pump, severely underestimating costs. The isothermal vacuum pump equation is replaced with an isentropic

one:

$$\dot{W}_{\text{pump, is}} = \frac{\gamma}{\gamma - 1} (\dot{n}_{\text{CO}_2, \text{out, L}} + \dot{n}_{\text{CO, out, L}}) R T \left[\left(\frac{p_{\text{atm}}}{p_L} \right)^{(\gamma-1)/\gamma} - 1 \right] \quad (3.32)$$

$$\dot{W}_{\text{pump, shaft}} = \frac{\dot{W}_{\text{pump, is}}}{\eta_{\text{pump}}} \quad (3.33)$$

With the ratio of specific heat constants $\gamma_{\text{mix}} = \frac{c_p}{c_v} = 1.30$, ignoring feed composition dependency on γ_{mix} and $\eta = 0.40$ as vacuum pumps are expected to be of lower efficiency.

The cost correlation equation is updated to reflect a liquid ring vacuum pump as per Turton (2012) [33]:

$$\log_{10}(C_p) = K_1 + K_2 \log_{10}(\dot{W}) + K_3 \left(\log_{10}(\dot{W}) \right)^2 \quad (3.34)$$

where C_p is the purchased equipment cost (USD, 2001) and $\dot{W}$ is the shaft power in kW. The installed cost and final fixed cost follow as:

$$C_{\text{installed}} = F_{BM} C_p \quad (3.35)$$

$$C_{\text{fixed}} = f_{\text{indirect}} C_{\text{installed}} \frac{\text{CEPCI}_{2024}}{\text{CEPCI}_{2001}} \quad (3.36)$$

With bare module factor $F_{BM} = 1.5$ and $f_{\text{indirect}} = 1.38$. The CEPCI values can be found in table A.4 in the appendix.

3.5.2 Kinetics Extension

The PSA model has two ways of adsorbing more CO_2 : 1) Increase the column allowing for more adsorbent area used for adsorption, or 2) increase the pressure difference between adsorption and desorption resulting in higher loadings and working capacities in a similar sized column.

Thus the PSA model is looking for a tradeoff between column size and compressor & pump duty. After correcting the inflated column CAPEX equation this tradeoff shifted materially in favor of column size. The new optimum found yielded extremely small working capacities that were physically unmeaningful. The inherent tradeoff the optimizer was missing is that at very small working capacities the kinetic effects heavily change and the adsorption and desorption speeds materially slow down, requiring even larger columns to compensate.

But crucially Tjahjana set ad- and desorption speeds to be independent of pressure, leading the optimizer to severely underestimate them.

To rectify this, the kinetics of the ad- and desorption process were expanded to be pressure dependent and considered individually as the model failed to correctly balance the tradeoff between higher pressure difference costs and slower kinetics after the corrections.

Besides the Sips isotherms, which were already implemented by Tjahjana into the model, Park et al.[29] also measured the kinetics:

Table 3.6: Non-isothermal kinetic parameters on pelletized zeolite 13X at 293 K

Component	Pressure (kPa)	$ha/\rho_s C_s$ (s ⁻¹)	β' (-)	D/R_P^2 (s ⁻¹)
CO ₂	11.95	0.012	0.163	0.0074
	29.47	0.011	0.263	0.0064
	43.23	0.011	0.336	0.0058
	59.71	0.013	0.474	0.0070
	87.33	0.012	0.606	0.0033
CO	8.040	0.021	0.068	0.0286
	28.72	0.019	0.157	0.0397
	46.33	0.020	0.226	0.0426
	59.83	0.027	0.400	0.0309
	77.78	0.029	0.448	0.0402

Characterized by equation (3.37), which is an asymptotic approximation that holds true for large t , for further details the reader is referred to Park et al. [29].

$$\frac{M_t}{M_\infty} = 1 - \frac{\beta'}{1 + \beta'} \exp\left(-\frac{ha}{\rho_s C_s} (1 + \beta')\right) \quad (3.37)$$

with M_t loading at time t , M_∞ equilibrium loading, $\frac{ha}{\rho_s C_s}$ overall heat-transfer coefficient, and correction factor β' .

Equation (3.37) is fit to the data for varying pressures and both species. To this end $\frac{ha}{\rho_s C_s}$ is averaged over all data points for a given species and β' is fit using nonlinear least squares:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n \left[\beta'(P_i) - \exp(\beta_0) * P_i^{\beta_1} \right]^2 \quad (3.38)$$

Table 3.7 and figure 3.3, figure 3.3 show the fit.

Table 3.7: Fitted non-isothermal kinetic parameters for CO₂ and CO on pelletized zeolite 13X at 293 K (NLS, linear β' scale)

Parameter	CO ₂	CO
$ha/\rho_s C_s$ mean (s ⁻¹)	0.0118	0.0232
β' slope β_1 (-)	0.7335	1.0411
β' intercept β_0 (-)	-3.7816	-5.3079
R^2 (linear scale)	0.988	0.951

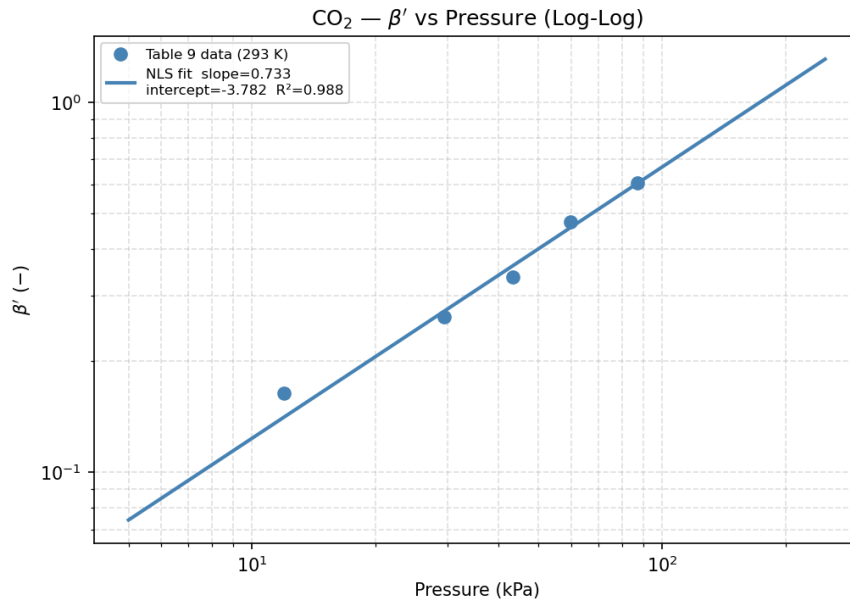


Figure 3.3: CO₂ — β' vs Pressure (Log-Log) at 293 K on pelletized zeolite 13X

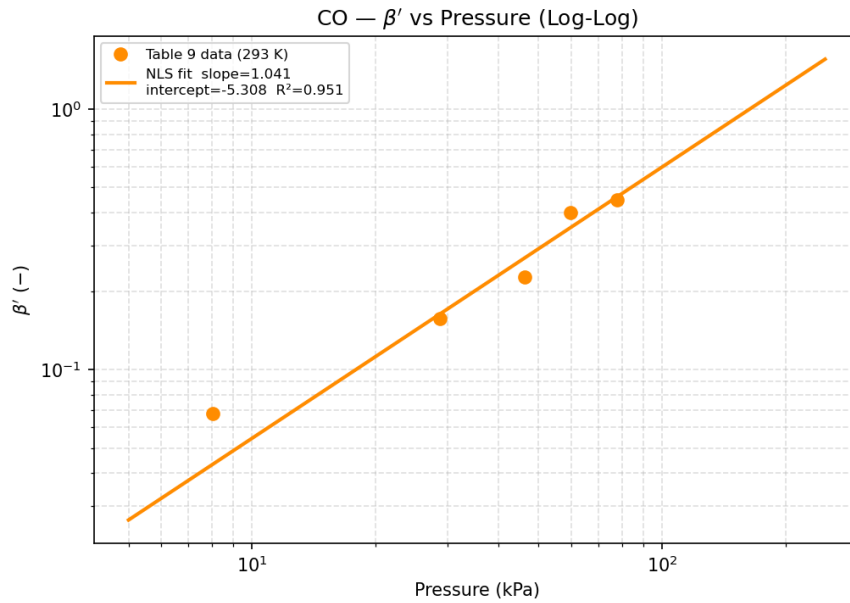


Figure 3.4: CO — β' vs Pressure (Log-Log) at 293 K on pelletized zeolite 13X

Figure 3.5 and figure 3.6 show the resulting uptake curves characterizing the loading efficiency $\frac{M_t}{M_\infty}$ over time. An extrapolated uptake curve 2 bar is also shown as the kinetics will have to be extrapolated due to lack of data.

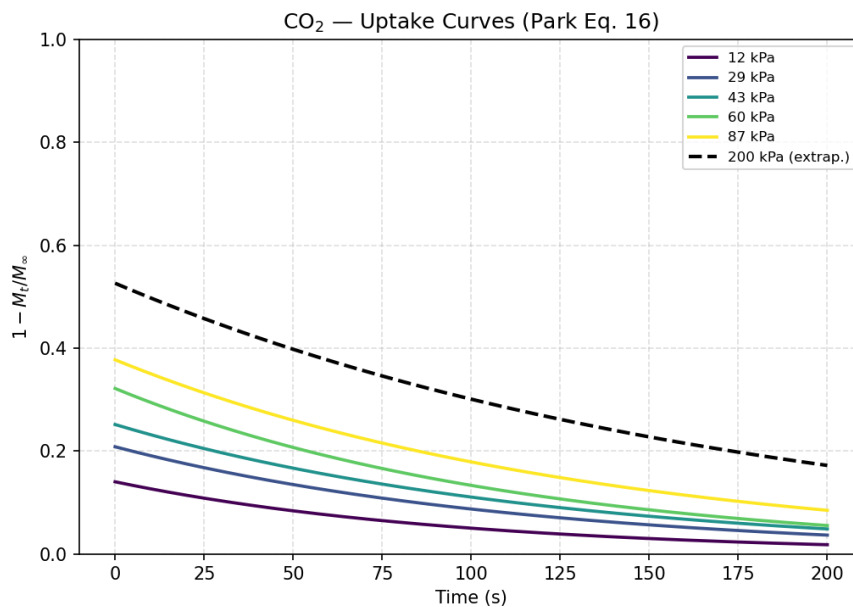


Figure 3.5: CO₂ — Uptake Curves at 293 K on pelletized zeolite 13X at five measured pressures with extrapolation to 200 kPa

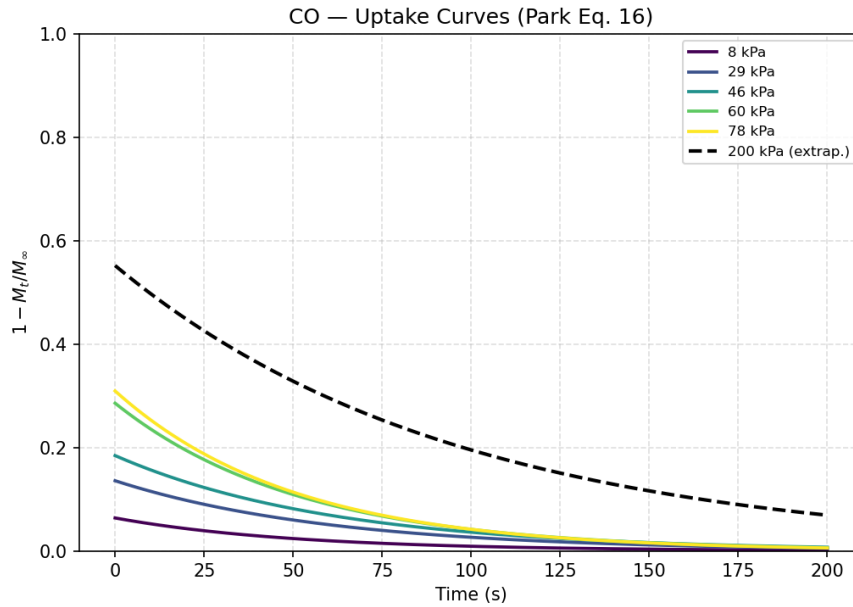


Figure 3.6: CO — Uptake Curves at 293 K on pelletized zeolite 13X at five measured pressures plus extrapolation to 200 kPa

3.5.3 Miscellaneous

The immediate consequences on the model are significantly increased pressure vessel volume requirements as a compensation for far slower kinetics as demonstrated in section 4.2.

Additional columns

The requirement for additional vessel volume coupled with an upper bound on vessel sizing of 100 ft height by 10 ft diameter necessitate the introduction of multiple columns N_{trains} . This inherently discrete variable is continuously relaxed for ease of solubility.

Inflation Adjustment

All equipment costs are inflation adjusted using the CEPCI [34].

Low-pressure column frozen solid approximation

Originally, the equilibrium loadings for the low pressure columns were calculated using.

$$q_{\text{CO}_2,L} = \frac{q_{\text{CO}_2}^s (B_{\text{CO}_2} y_{\text{CO}_2,\text{des}} p_L)^{1/n_{\text{CO}_2}}}{1 + (B_{\text{CO}_2} y_{\text{CO}_2,\text{des}} p_L)^{1/n_{\text{CO}_2}} + (B_{\text{CO}} y_{\text{CO},\text{des}} p_L)^{1/n_{\text{CO}}}} \quad (3.39)$$

$$q_{\text{CO},L} = \frac{q_{\text{CO}}^s (B_{\text{CO}} y_{\text{CO},\text{des}} p_L)^{1/n_{\text{CO}}}}{1 + (B_{\text{CO}_2} y_{\text{CO}_2,\text{des}} p_L)^{1/n_{\text{CO}_2}} + (B_{\text{CO}} y_{\text{CO},\text{des}} p_L)^{1/n_{\text{CO}}}} \quad (3.40)$$

With $q_{i,L}$ equilibrium loading, Sips constants q_i^s and $B_i 1/n_i$, pressure p_L and desorption mole fraction $y_{i,\text{des}}$.

These equations link the equilibrium loadings to the composition at the outlet. In reality, the equilibrium loadings change over the runtime of the desorption process as the concentrations of the outlet slowly change. This relationship can naturally not be modeled using a steady-state model but the approach used above gives overly optimistic equilibrium loadings leading to higher working capacities and smaller columns.

Sees et al.[16], which proposed the underlying virtual moving bed framework Tjahjaja built the PSA on, mentions this exact issue and proposes to use the frozen-solid approach where the equilibrium loadings are linked to the feed composition instead to get a more physically meaningful working capacity:

$$q_{\text{CO}_2,L} = \frac{q_{\text{CO}_2}^s (B_{\text{CO}_2} y_{\text{CO}_2,\text{in}} p_L)^{1/n_{\text{CO}_2}}}{1 + (B_{\text{CO}_2} y_{\text{CO}_2,\text{in}} p_L)^{1/n_{\text{CO}_2}} + (B_{\text{CO}} y_{\text{CO},\text{in}} p_L)^{1/n_{\text{CO}}}} \quad (3.41)$$

$$q_{\text{CO},L} = \frac{q_{\text{CO}}^s (B_{\text{CO}} y_{\text{CO},\text{in}} p_L)^{1/n_{\text{CO}}}}{1 + (B_{\text{CO}_2} y_{\text{CO}_2,\text{in}} p_L)^{1/n_{\text{CO}_2}} + (B_{\text{CO}} y_{\text{CO},\text{in}} p_L)^{1/n_{\text{CO}}}} \quad (3.42)$$

Subsequently, this new formulation for the equilibrium loadings is implemented in accordance with Sees et al.[16].

3.6 Staged Compression

Compression costs contribute significantly to membrane separation costs. Hesselmann et al. [23] ignored the Joule-Thompson Effect. In order to model the requirement to cool after compression, and the worse compression performance from higher temperature gas, a staged isentropic compression model with intercooler after every stage is created.

3.6.1 Compressor Model

For each compression stage $k \in \{1, \dots, N\}$, the isentropic outlet temperature is computed from

$$T_{out,k} = T_{in,k} + \frac{T_{in,k}}{\eta} \left[\left(\frac{P_{out,k}}{P_{in,k}} \right)^{(\gamma-1)/\gamma} - 1 \right], \quad (3.43)$$

where $T_{in,k}$ is the stage inlet temperature (K), $P_{in,k}$ and $P_{out,k}$ are the inlet and outlet pressures (bar), $\gamma = 1.3$ is the isentropic heat capacity ratio, and $\eta = 0.9$ is the isentropic efficiency. The shaft power demand of stage k is

$$\dot{W}_k = \dot{n} c_{p,mix} (T_{out,k} - T_{in,k}), \quad (3.44)$$

where $\dot{n}$ (mol/s) is the molar feed flow rate and $c_{p,mix}$ (J/mol·K) is the molar heat capacity of the mixture evaluated at the stage inlet temperature via the Shomate equation (see below).

The compressor purchase cost follows the Guthrie [30] correlation, adjusted to 2024 prices via the Chemical Engineering Plant Cost Index (CEPCI):

$$C_{base} = \omega_1 \left(\frac{\dot{W}_k}{\omega_2} \right)^{\omega_3}, \quad (3.45)$$

$$C_{Fa} = \omega_4 C_{base}^{\omega_5}, \quad (3.46)$$

$$C_{inst} = \omega_6 (C_{base} + C_{Fa}), \quad (3.47)$$

$$C_{CAPEX, comp} = \omega_7 \omega_8 C_{inst} \frac{CEPCI_{2024}}{CEPCI_{1974}}, \quad (3.48)$$

with constants ω_1 – ω_8 given in table A.5. The CEPCI values can be found in table A.4 in the appendix. The annual operating cost combines electricity consumption and maintenance:

$$\dot{C}_{OPEX, comp} = \dot{W}_k h_{op} c_{elec} + f_{maint, rot} C_{CAPEX, comp}, \quad (3.49)$$

with h_{op} (hr/yr) operating hours, c_{elec} (€/kWh) electricity price, and $f_{maint, rot} = 0.05$ [35].

3.6.2 Heat Exchanger Model

For each heat exchanger the thermal duty is

$$\dot{Q} = \dot{n} \Delta h_{mix} \quad (3.50)$$

where the difference in enthalpy is $\Delta h_{mix} = h_{in}(T_{in}) - h_{out}(T_{out})$. The required heat transfer area is derived from the overall heat transfer coefficient U and log-mean temperature difference ΔT_{lm} :

$$A = \frac{\dot{Q}}{U \Delta T_{lm}}, \quad (3.51)$$

where $U = 850 \text{ W/m}^2\text{K}$ and $\Delta T_{lm} = 20 \text{ K}$ for water cooling. The purchased cost follows from the Guthrie [30] shell-and-tube correlation:

$$C_{p,HX} = 101.3 (A_{ft^2})^{0.65}, \quad (3.52)$$

escalated and installed via

$$C_{\text{CAPEX},HX} = C_{p,HX} \frac{CEPCI_{2024}}{CEPCI_{1969}} F_{BM} f_{ind} r_{USD/EUR}, \quad (3.53)$$

with bare module factor $F_{BM} = 3.29$ (Turton et al., 2012). Annual OPEX covers cooling utility cost and maintenance:

$$\dot{C}_{\text{OPEX},HX} = \dot{Q} h_{op} c_{cool} + f_{maint} C_{\text{CAPEX},HX}, \quad (3.54)$$

where c_{cool} is the medium-specific cooling utility price ($0.00131 \text{ EUR/kWh}_{th}$ for water cooling [36]) and $f_{maint} = 0.03$ [35] (static equipment).

3.6.3 Mixture Heat Capacity, Shomate Equation

Equation (3.44) and equation (3.50) require the specific heat capacity of the mixture and enthalpy of mixture respectively.

Assuming Ideal Gas and Ideal Mixing Rule they can be approximated using the Shomate Equation:

Pure-component molar heat capacities are evaluated using the NIST Shomate equation [37]:

$$C_{p,i}(T) = A_i + B_i t + C_i t^2 + D_i t^3 + \frac{E_i}{t^2}, \quad t = \frac{T}{1000}, \quad (3.55)$$

where the Shomate coefficients $(A_i, B_i, C_i, D_i, E_i)$ for each component $i \in \{CO_2, CO, H_2\}$ are taken from the NIST WebBook [37]. The mixture heat capacity follows the ideal mixing rule:

$$c_{p,mix}(T) = \sum_i z_i C_{p,i}(T), \quad (3.56)$$

where z_i is the mole fraction of component i . The mixture heat capacity is a linear combination of the pure heat capacities scaled by their mole fractions as per the ideal mixing rule. The molar enthalpy of component i at temperature T relative to the standard state at 298.15 K is

$$H_i(T) = A_i t + \frac{B_i}{2} t^2 + \frac{C_i}{3} t^3 + \frac{D_i}{4} t^4 - \frac{E_i}{t} + F_i - H_i^\circ, \quad (3.57)$$

and the mixture enthalpy change across a heat exchanger spanning temperatures T_{hot} to T_{cold} is

$$\Delta h_{mix} = \sum_i z_i [H_i(T_{hot}) - H_i(T_{cold})]. \quad (3.58)$$

3.7 Amine Washing

To meet a target syngas rate of 75t/day, the CO₂ electrolyzer produces ≈ 28 mol/s of CO . Typically the electrolyzer runs in large excess of CO₂ , together with a couple of moles of H₂ produces this nets a total mole outflow of just under 100 mol/s.

The amine washing model will see that same mole flow. The table below shows the amine washing techno-economic surrogate model landscape available.

Table 3.8: Modelling ranges of CO₂ capture surrogate models across three studies

Study	Solvent	Capture Rate (–)	Inlet Flow (mol/s)	Inlet CO ₂ (mol/mol dry)
Chung & Lee [38]	MEA	0.20–0.98	100–10,000	0.03–0.50
	PZ	0.20–0.98	100–10,000	0.03–0.50
Kim & Léonard [39]	MEA	0.90 (fixed)	≈ 1 –1,614	0.05–0.50
Hsiao & Chang [40]	MEA	not fixed	≈ 0.19 –0.62	0.05–0.15

The only model that meets the mole flow target range for our case is Kim & Léonard [39]. Unfortunately the outflow CO₂ mass fraction is capped at 50 % of total flow and the CO₂ capture rate is fixed at 90 % which will mean a syngas flow that does not meet the 98 % purity standard that the PSA and membrane flowsheets achieve. Nevertheless this model is chosen as it is the only available option.

Kim & Léonard created a regression to determine the total equipment cost (TEC):

$$\text{TEC} = \alpha_{\text{TEC}} + (\beta_{\text{TEC}} y_{\text{CO}_2, \text{in}}^{n_{\text{TEC}}} + \gamma_{\text{TEC}}) F^{m_{\text{TEC}}} \quad (3.59)$$

where F is the volumetric feed-gas flow rate (in 10³ Nm³/h) and $y_{\text{CO}_2, \text{in}}$ is the CO₂ mole fraction of the feed. The remaining coefficients are taken from the regression of Kim & Léonard [39] and listed in table A.6.

An installation factor $f_{\text{inst}} = 3.8$ as well as a factor 2 for direct and indirect cost, as detailed by Guthrie [30] are added to get the final CAPEX.

$$C_{\text{CAPEX, amine}} = 2 f_{\text{inst}} \text{TEC} \quad (3.60)$$

The OPEX equation incorporates 5 cost sources: the reboiler duty $\dot{Q}_{\text{reb}}$, electricity $\dot{W}_{\text{elec}}$, cooling duty $\dot{Q}_{\text{cool}}$:

$$\dot{C}_{\text{OPEX, amine}} = \dot{Q}_{\text{reb}} c_{\text{conv}} c_{\text{steam}} + \dot{W}_{\text{elec}} c_{\text{elec}} + \dot{Q}_{\text{cool}} c_{\text{conv}} c_{\text{cw}} + c_{\text{MEA}} c_{\text{conv}} \dot{n}_{\text{gas, in}} + \frac{f_{\text{maint}} C_{\text{CAPEX, amine}}}{h_{\text{op}}} \quad (3.61)$$

The individual duties follow surrogate equations:

$$\dot{Q}_{\text{reb}} = \dot{m}_{\text{CO}_2, \text{cap}} (\alpha_{1,1} e^{\alpha_{1,2} y_{\text{CO}_2}} + \alpha_{1,3} e^{\alpha_{1,4} y_{\text{CO}_2}}) \quad (3.62)$$

$$\dot{W}_{\text{elec}} = \dot{m}_{\text{CO}_2, \text{cap}} (\alpha_{2,1} e^{\alpha_{2,2} y_{\text{CO}_2}} + \alpha_{2,3} e^{\alpha_{2,4} y_{\text{CO}_2}}) \quad (3.63)$$

$$\dot{Q}_{\text{cool}} = \dot{m}_{\text{CO}_2, \text{cap}} (\alpha_{3,1} e^{\alpha_{3,2} y_{\text{CO}_2}} + \alpha_{3,3} e^{\alpha_{3,4} y_{\text{CO}_2}}) \quad (3.64)$$

with the captured- CO_2 mass rate $\dot{m}_{\text{CO}_2, \text{cap}}$ (in t/h), the CO_2 mole fraction y_{CO_2} , and the surrogate parameters $\alpha_{i,j}$ listed in table A.7.

3.8 Economic Parameters and Solver

The key economic parameters are listed in table 3.9

CO_2 feed cost, electricity cost, currency conversion, operating hours and plant life are taken from [24]. The interest rate is taken from the amine wash surrogate [39] and kept consistent across all submodels. The maintenance factors are from Guthrie [30] and finally the steam cost is linked to the electricity cost with a boiler efficiency of 95%:

Table 3.9: Economic Parameters

Economic Parameter	Value
CO ₂ Feed Cost	22 €/t
Electricity Cost	0.08 €/kWh
Steam Cost	23.4 €/GJ
Currency Conversion	0.96 \$/€
Operating Hours	8400 h/yr
Plant Life	20 yr
Maintenance Factor Rotating	5%
Maintenance Factor Static	3%
Interest Rate	8.5%

$$c_{\text{steam}} = \frac{c_{\text{elec}}}{\eta_{\text{boiler}}} \quad (3.65)$$

The interior point optimizer (IPOPT) is used to solve all flowsheets. As it is a local optimizer a multistart approach with 20 multistarts are used.

4 Sensitivity Analysis of the CO₂ Electrolysis Surrogate and PSA

4.1 CO₂ Electrolysis Surrogate

A sensitivity analysis is performed around the operating point $E_{appl} = -1.25\text{ V}$, $v_{gas} = 27.5\text{ ml/min}$, $CO_{2,share} = 0.8$ to compare the fitted equations with the base model. Every input variable is plotted against every output concentration. Only the more interesting plots are shown in this chapter; for the full list, the reader is referred to section A.7.

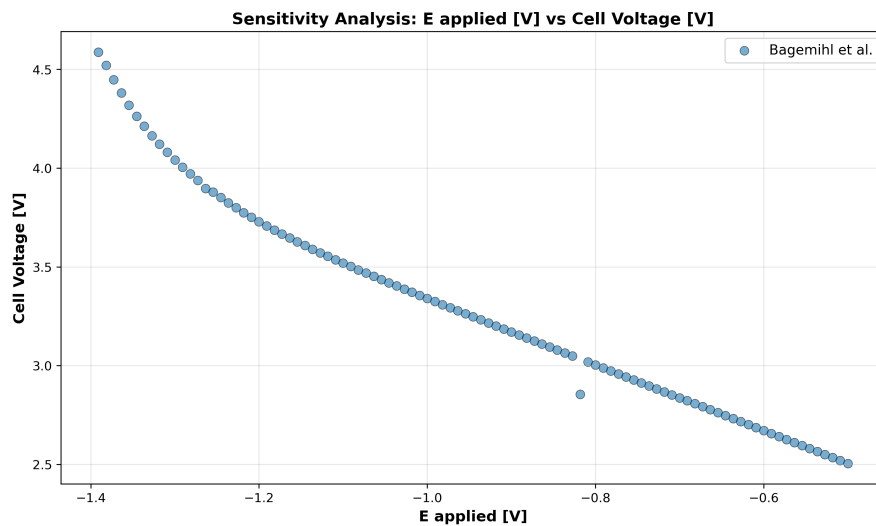


Figure 4.1: Sensitivity analysis: E_{appl} vs Cell Voltage

Figure 4.1 shows the relationship between the applied voltage and the cell voltage as described by equation (3.10) - equation (3.16). The relationship is retrieved from converged operating points which explains the outlier at $E_{appl} \approx -0.82$.

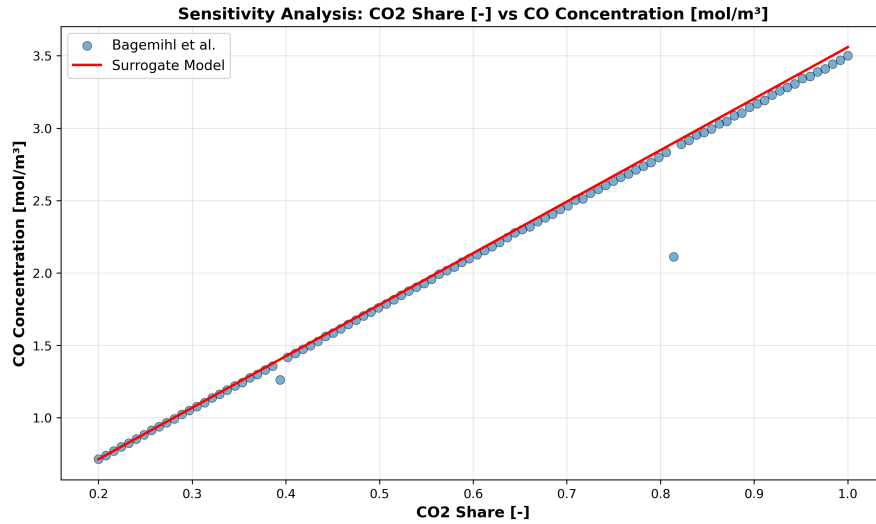


Figure 4.2: Sensitivity analysis: Effect of CO₂ share on CO concentration output at $E_{\text{appl}} = -1.25 \text{ V}$ and $v_{\text{gas}} = 27.5 \text{ ml/min}$

Figure 4.2 shows how the CO output scales with the CO₂ share in the feed. The increased amount of CO₂ helps drive the electrolysis reaction, so the CO concentration grows linearly. The two converged points not following the linear trend are deemed outliers.

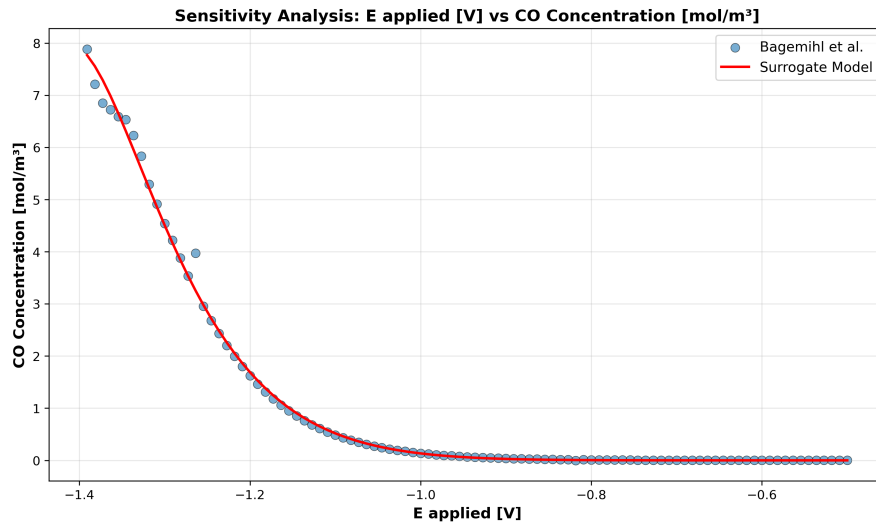


Figure 4.3: Sensitivity analysis: Effect of applied voltage E_{appl} on CO concentration output at $\text{CO}_2_{\text{share}} = 0.8$ and $v_{\text{gas}} = 27.5 \text{ ml/min}$

The applied voltage naturally has the biggest effect on the CO production. Figure 4.3 shows that increasing the voltage raises the CO concentration steeply. This is the

expected behaviour, as larger voltages drive the CO₂ reduction reaction harder. Convergence for the Bagemihl et al. model generally gets harder and more unstable at higher voltages.

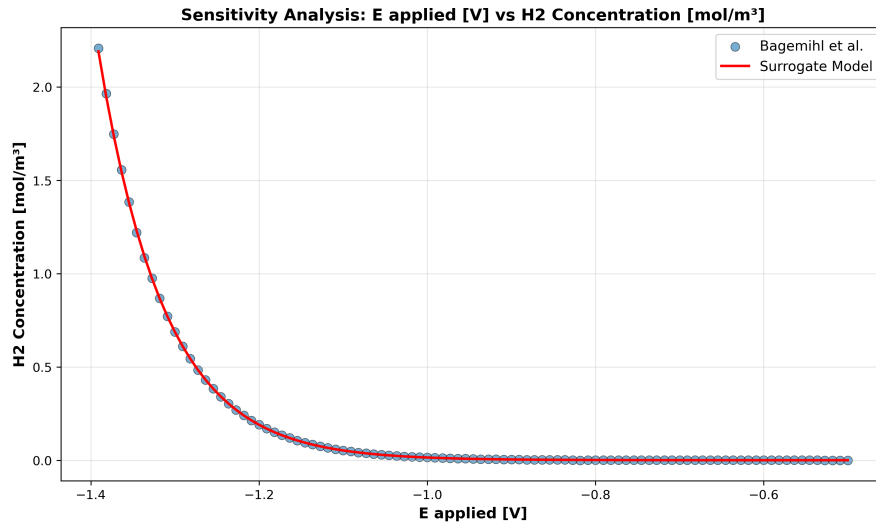


Figure 4.4: Sensitivity analysis: Effect of applied voltage E_{appl} on H₂ concentration output at $CO_{2\text{share}} = 0.8$ and $v_{\text{gas}} = 27.5$ ml/min

Hydrogen is produced as a side product through the competing hydrogen evolution reaction. Figure 4.4 shows the H₂ concentration increasing at higher voltages. Those operating points pay an increasing selectivity penalty toward H₂.

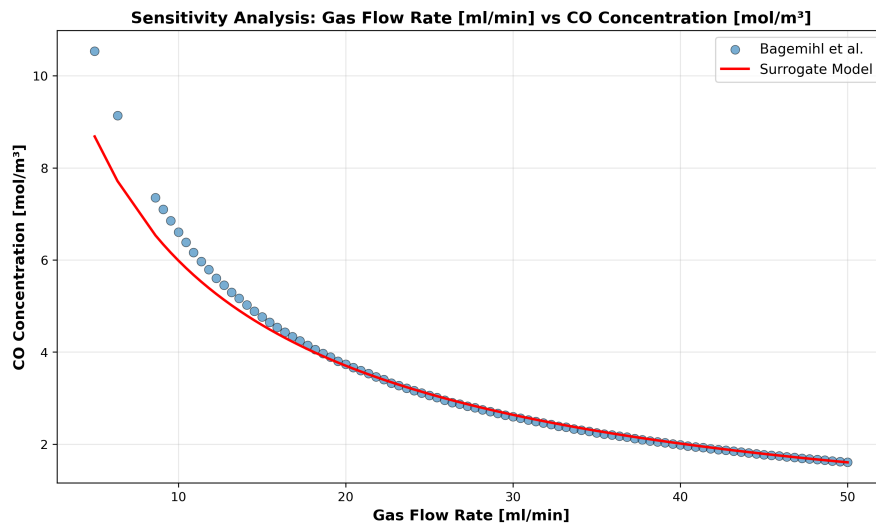


Figure 4.5: Sensitivity analysis: Effect of gas flow rate v_{gas} on CO concentration output at $E_{\text{appl}} = -1.25$ V and $CO_{2\text{share}} = 0.8$

Figure 4.5 shows that increasing the flow lowers the CO concentration, since each CO₂ molecule spends less time at the electrode and a smaller fraction is converted.

4.2 Pressure Swing Adsorption

To get a sense for the inner workings and tradeoffs of the new PSA model with kinetics included a parametric sweep is done.

To this end the inlet is fixed to be 66 % CO₂ and 33 % CO at a total mole flow of 85 mol/s. The of the CO rich product gas is set to 98 %. The adsorption pressure of the PSA is fixed from 1 - 5 bar and for each of those points the total lifecycle cost of 20 years is minimized. The optimizer chooses the inner workings, including the desorption pressure accordingly.

Inner Workings of the PSA Model

The figures below show the optimal operating point (if any) the optimizer found.

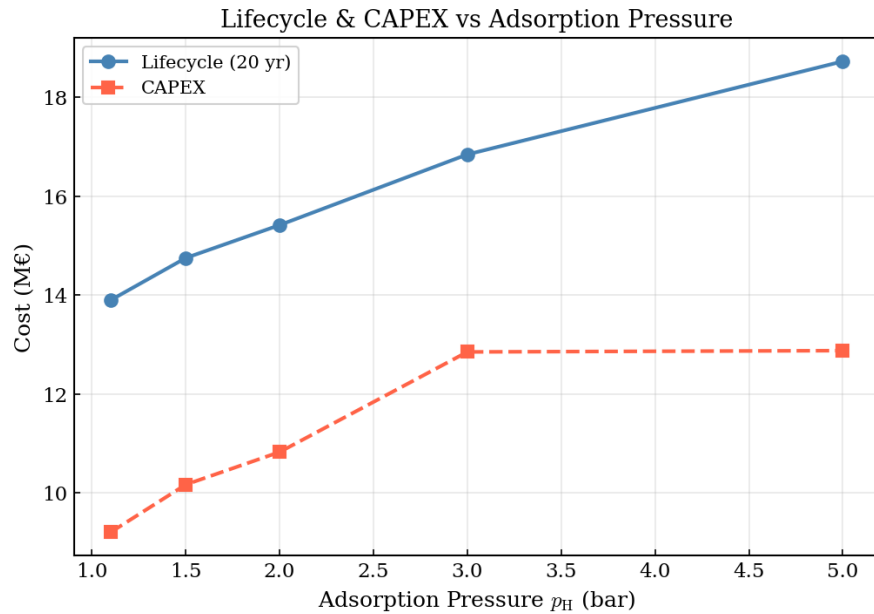


Figure 4.6: PSA: Lifecycle & CAPEX vs Adsorption Pressure

Figure 4.6 shows the lower the adsorption pressure, the better the economics. Lifecycle of the 20 year lifespan is the lowest at ~ 1 bar. Since the CAPEX is also increasing

with increasing adsorption pressure the logical conclusion is the optimizer finds the OPEX of the vacuum pump together with larger columns to be more cost effective than simply running the compressor at higher pressure.

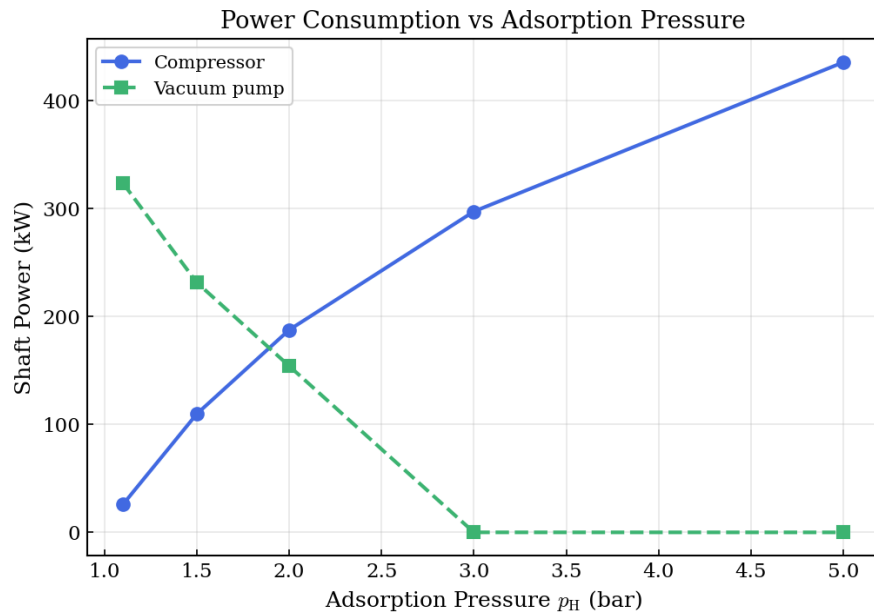


Figure 4.7: PSA: Shaft Power of Vacuum Pump & Compressor vs Adsorption Pressure

Indeed as figure 4.7 shows the lower the adsorption pressure the higher the vacuum pump power needed to keep a decent working capacity between the ad- and desorption columns.

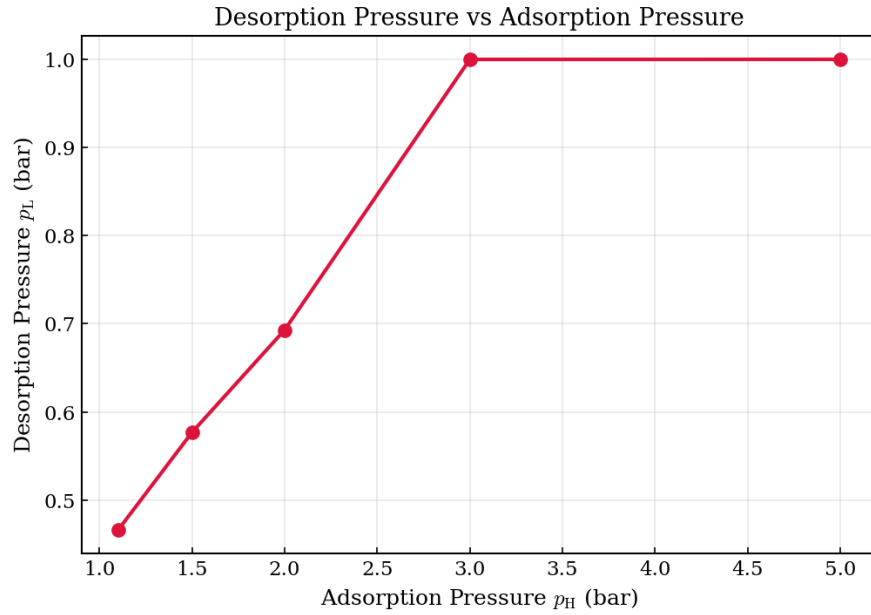


Figure 4.8: PSA: Desorption Pressure vs Adsorption Pressure

As expected to keep the working capacity going the desorption pressure has to be far enough from the adsorption pressure as shown in figure 4.8.

For more plots on the PSA's internal operating variables, including cycle time, adsorbent mass, bed saturation, the equilibrium CO₂ and CO loadings and the working capacities, the reader is referred to section A.8.

CO Recovery Limitation

An important caveat about this model is the rather low CO recovery as can be seen in figure 4.9.

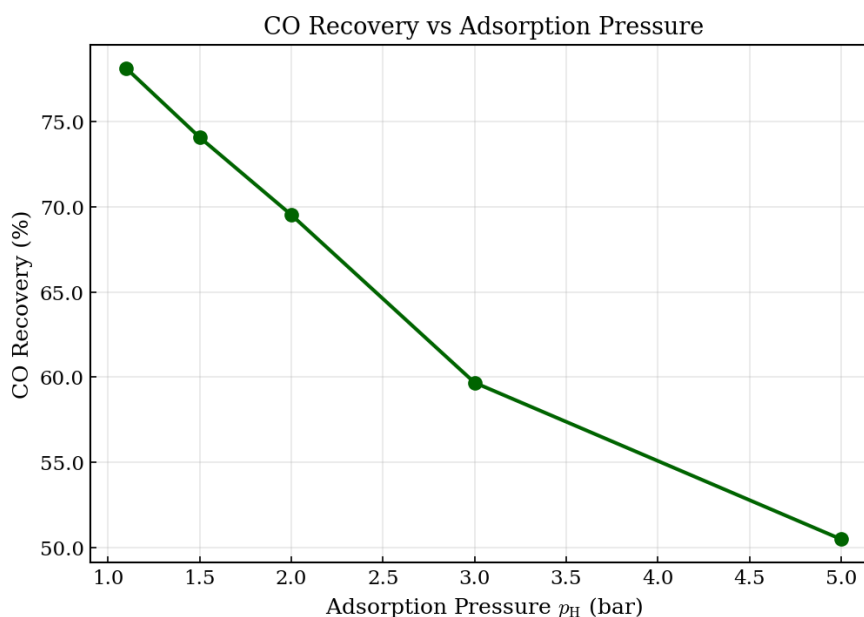


Figure 4.9: PSA: CO Recovery vs Adsorption Pressure

At best, around 80 % of the CO supplied by the feed ends up in the product stream, with the remainder going to the recycle. This is a natural and expected consequence of the two-step model chosen. At the end of the adsorption phase, part of the CO is adsorbed onto the zeolite and part remains in the void space between the solid particles. When the model switches to desorption, all of this CO is pushed into the recycle stream. To mitigate this, real PSA units insert a rinse phase between the adsorption and desorption phases: a small portion of the purge/recycle stream from the previous cycle or an adjacent column is taken, repressurized, and fed back into the column co-currently. This rinse, consisting mostly of CO₂, displaces the CO remaining in the void space out of the column and outcompetes some of the CO adsorbed on the zeolite. Both of these additional CO sources reach the product stream, thereby improving recovery.

5 Flowsheet Optimization Results

This chapter discusses and compares the optimal operating point for the three flowsheet alternatives in the base case in section 5.1 and after varying key economic parameters in section 5.2.

5.1 Base Case

This section compares the optimal operating point and its costs for all three flowsheets with economic parameters set as mentioned in table 3.9 in section 3.8.

All flowsheets produce 75 t/day of syngas at a syngas ratio of 2:1. The PSA and membrane flowsheets produce syngas at a purity of at least 98%. In the amine wash flowsheet the CO₂ capture rate is fixed at 90% as per model specifications.

5.1.1 Economic Results

Figure 5.1 shows the syngas production cost for each flowsheet and their distribution of CAPEX and OPEX.

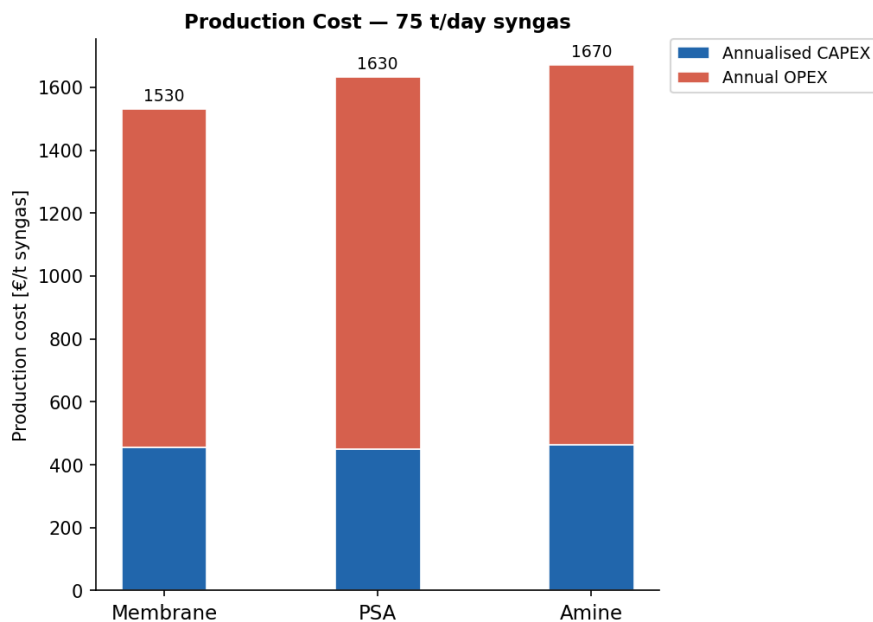


Figure 5.1: Base Case: Syngas Production Cost

In the base case membrane separation is the cheapest, followed by the PSA and amine wash. Part of the gap to the membrane flowsheet stems from a lower CO capture rate in the PSA unit, which lets more product slip and raises the upstream load; this is examined further in section 5.1.3.

Figure 5.2 shows the breakdown of the OPEX for each flowsheet.

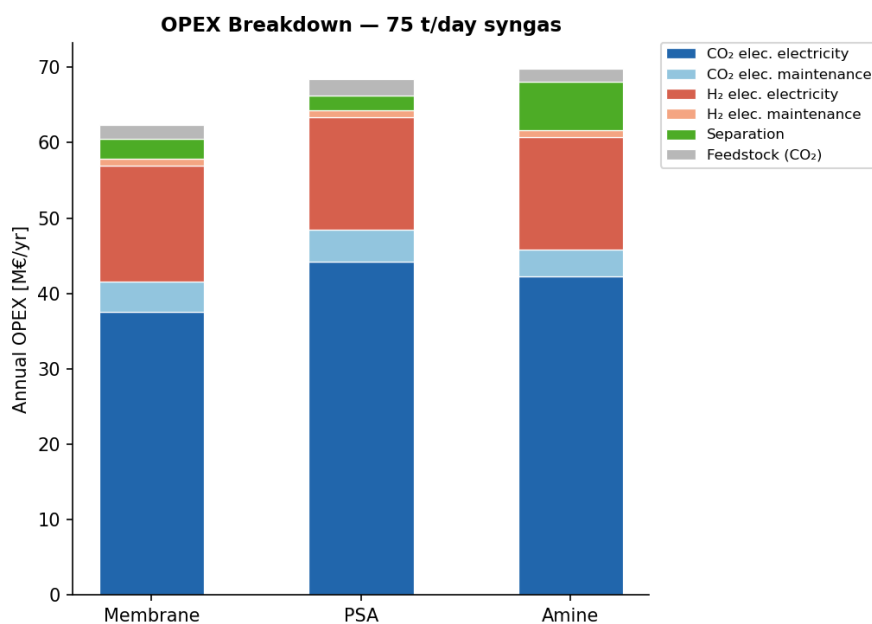


Figure 5.2: Base Case: OPEX Breakdown

OPEX is largely dominated by CO₂ electrolyzer electricity cost. The H₂ electrolysis electricity cost is constant across all three flowsheets since in all cases the H₂ production from the CO₂ electrolyzer is negligible.

Amine wash has the highest separation-unit OPEX, as its feed to the separation is rich in CO₂ and thus the reboiler duty for the amine wash becomes large. CO₂ electrolyzer electricity cost and feedstock cost vary slightly depending on the separation unit used.

Figure 5.3 breaks down the CAPEX for each flowsheet.

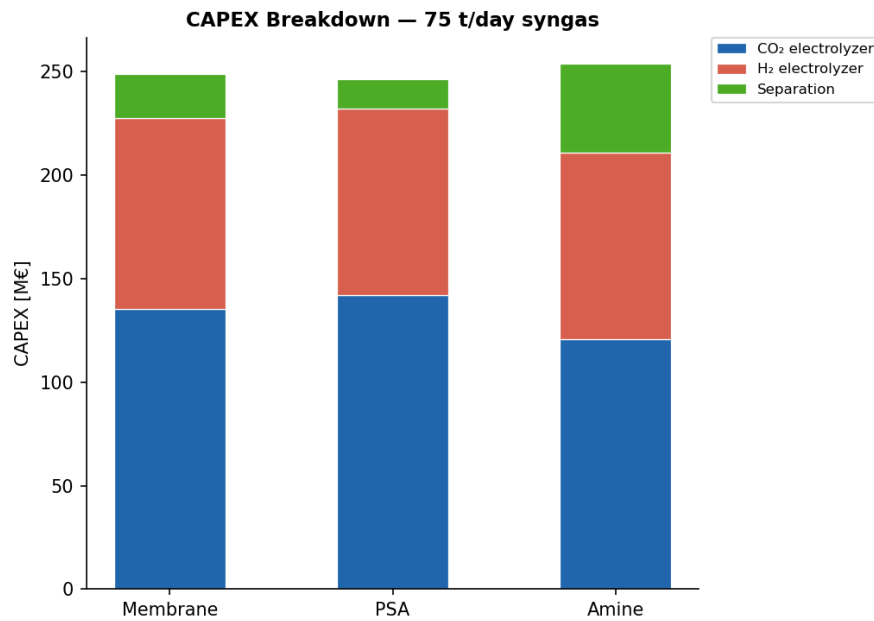


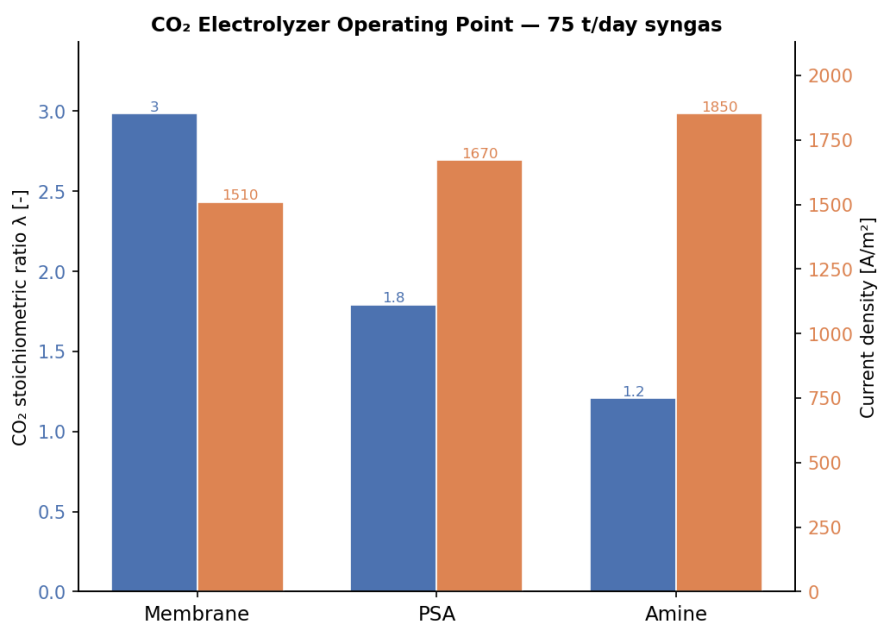
Figure 5.3: Base Case: CAPEX Breakdown

CAPEX of the separation unit is most pronounced for amine wash, followed by membrane and finally PSA. The H₂ electrolysis cost is again constant, as in the OPEX, since the requirements on the electrolysis unit are constant across all three flowsheets. The CAPEX of the CO₂ electrolyzer varies slightly, suggesting a small difference in operating point.

5.1.2 Operating Point

This section examines key variables of the operating point of all three flowsheets.

Figure 5.4 provides deeper insight into the CO₂ electrolysis operating point. It shows the stoichiometric ratio λ and current density as defined by equation (2.10) and equation (2.11).

Figure 5.4: Base Case: CO₂ Excess & Current Density

The stoichiometric ratio λ varies strongly across the flowsheets: the membrane flow-sheet favors a higher ratio at lower current density compared with PSA and amine wash.

Figure 5.5 shows the total homogeneous consumption and heterogeneous conversion as defined by equation (2.7) and equation (2.8).

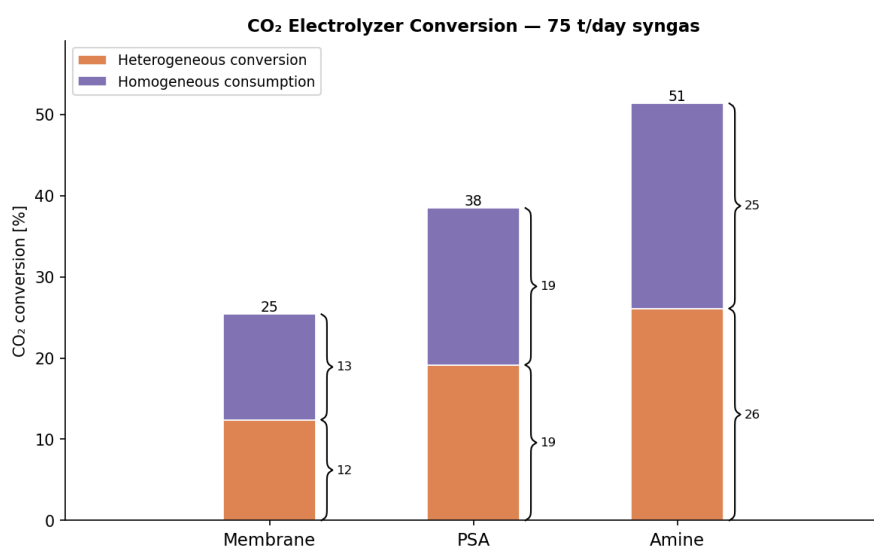


Figure 5.5: Base Case: Heterogeneous Conversion & Homogeneous Consumption

Total conversion, i.e., homogeneous consumption plus heterogeneous conversion, is smallest for the membrane flowsheet, in line with the increased CO₂ stoichiometric ratio λ shown in figure 5.4.

Homogeneous consumption and heterogeneous conversion split roughly 50/50 in each case, meaning half the reacted CO₂ gets wasted into the electrolyte. This large waste stream is a known issue in CO₂ electrolysis and Bagemihl's M3 model [10] (the source for the CO₂ electrolyzer model of this work) showed the same behavior.

Table 5.1: Key operating-point parameters of the membrane separation unit (75 t/day syngas).

Flowsheet	Parameter	Value
Membrane	CO ₂ Inlet Mole Fraction	0.77
	Feed (Retentate) Pressure	18 bar
	Permeate Pressure	1.0 bar
	CO Recovery	93%
	Syngas Purity	99%

The membrane permeate pressure is fixed at 1 bar and the feed pressure is set by the optimizer to 18 bar. A two-stage compression train is used to keep compressor temperatures at manageable levels (< 200 °C). The CO₂ mole fraction at the CO₂ electrolyzer outlet and thus membrane inlet is high at 0.77. The syngas product purity is 99% and 93% of the CO is sent to the product stream with only 7% going back into the recycle.

Table 5.2: Key operating-point parameters of the PSA separation unit (75 t/day syngas).

Flowsheet	Parameter	Value
PSA	CO ₂ Inlet Mole Fraction	0.76
	Adsorption Pressure p_H	100 kPa
	Desorption Pressure p_L	10 kPa
	CO Recovery	86%
	Syngas Purity	99%

The CO₂ inlet mole fraction and product purity of the PSA flowsheet are similar to the membrane case at 0.76 & 99% respectively. The optimizer sets ad- and desorption pressures to their lower bounds and achieves a CO recovery of 86%, i.e., 14% of CO get sent back to the recycle and through the CO₂ electrolyzer.

Table 5.3: Key operating-point parameters of the amine separation unit (75 t/day syngas). Cost shares are fractions of the amine separation-unit OPEX.

Flowsheet	Parameter	Value
Amine	CO ₂ Inlet Mole Fraction	0.38
	CO ₂ Capture Rate	90%
	Reboiler Cost Share of OPEX	78%
	Maintenance Cost Share of OPEX	20%
	Cooling, Electricity & Makeup Share of OPEX	0.7%
	Syngas Purity	94%

The CO₂ inlet mole fraction of the amine wash flowsheet is considerably lower at 0.38 and does not hit the upper bound of 0.5 set by the amine wash model. The amine wash model fixes the CO₂ capture rate at 90% which entails a syngas purity of 94%. The amine wash OPEX is dominated by reboiler duty and maintenance.

5.1.3 Discussion

Syngas production costs are dominated by OPEX and thus by extension by electrolyzer power draw. The H₂ electrolyzer produces an essentially fixed amount of H₂ and is thus the biggest lever left for reducing syngas production cost is increasing CO₂ electrolyzer efficiency at comparatively cheap separation cost.

Amine wash is the most expensive separation option, both on the flowsheet level and at the individual separation level in CAPEX and OPEX. The flowsheet doesn't meet purity requirements due to model limitations. Increasing purity would presumably increase cost even more, widening the cost gap to the PSA and membrane flowsheets. The CO₂ inlet mole fraction is comparatively low since reboiler OPEX costs are pronounced and the CO₂ stoichiometric ratio λ is set almost to unity to reduce amine wash costs but at the cost of lower electrolyzer efficiency.

The membrane flowsheet is the cheapest option. Its moderate recovery rate, coupled with cheap OPEX and CAPEX, allows the electrolyzer to efficiently run at higher CO₂ stoichiometric ratio λ and highly pure inlet conditions. These operating conditions allow for the least power-hungry CO₂ electrolysis unit which is the main determining factor in the production cost.

The PSA flowsheet is the middle option, even though the separation unit itself is the cheapest in OPEX and CAPEX. The poorer CO recovery performance forces the CO₂ electrolysis unit to be larger and more energy intensive. This makes the PSA flowsheet the most expensive of the three in both CAPEX and OPEX and raises production cost accordingly. The CO recovery is pushed as high as possible, as evidenced by the ad- and desorption pressures sitting at their lower bounds. As discussed in section 4.2 this is partly a model issue because the current PSA model only considers an ad- and desorption phase but is missing an intermediate blowdown phase typically used to increase product recovery. Such a model improvement could make the PSA separation competitive with the membrane flowsheet.

5.2 Sensitivity Analysis

Figure 5.6 shows a sensitivity analysis for all three flowsheets. Electricity price is a main cost contributor. Due to its early stage development, CO₂ electrolyzer area cost is still uncertain. Both electricity price and area cost are varied by 50%. CO₂ price is varied by 100% to 0 €/t, accounting for possible carbon credit crediting of the CO₂. The gray bars denote the production cost of the base case as discussed in the previous chapter.

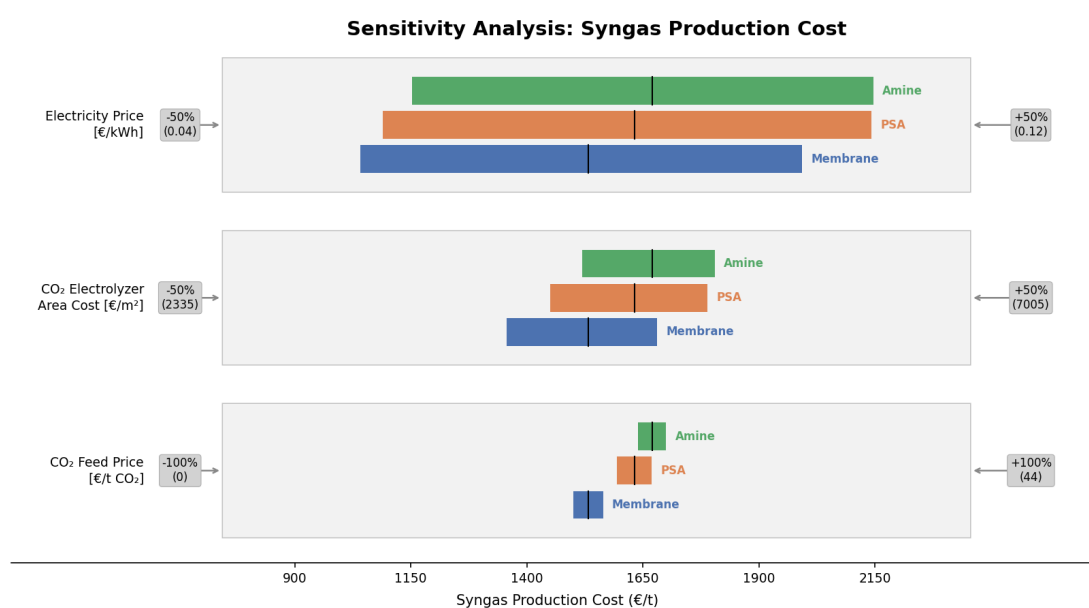


Figure 5.6: Sensitivity Analysis of Flowsheets

Electricity price has the largest impact on syngas production cost. At 0.12 €/kWh the PSA and amine flowsheets are equally expensive while at low prices the gap between PSA and membrane narrows. PSA is the most sensitive to electricity price due to its energy inefficient vacuum pump.

The CO₂ electrolyzer area cost also has a moderate influence on production cost while keeping the ranking between the separation alternatives constant.

The CO₂ feed price has a minor influence on syngas production cost while not radically changing the economics, even at 0 €/t.

6 Summary and Outlook

The goal of this work is a techno-economic comparison between three alternative separation units (pressure-swing adsorption, membrane separation & amine washing) for the production of green syngas through CO₂ electrolysis. To this end three corresponding flowsheets were built for a fixed target production of 75 t/day of syngas at a 2:1 H₂:CO ratio. The lifecycle cost of each flowsheet was minimized, finding an optimal operating point for the separation units and electrolyzer in the process.

Each flowsheet is composed of several submodels. To keep the flowsheet optimization tractable a CO₂ electrolysis surrogate was created. Additional submodels for the membrane and an amine separation, as well as a model for the H₂ electrolysis were adapted from literature. Finally an existing pressure-swing adsorption model was corrected and its kinetics extended to be pressure dependent.

The membrane separation was found to be the cheapest (1530 €/t syngas), followed by pressure-swing adsorption (1630 €/t syngas) and finally amine washing (1670 €/t syngas). The process is OPEX dominated, with CO₂ electrolysis electricity cost being the main cost contributor. While the separation units themselves are cheap compared to both electrolysis units in OPEX & CAPEX, they have a significant effect on CO₂ electrolysis OPEX through the recycle. High purity in the CO₂ recycle coupled with cheap separation cost allows the CO₂ electrolysis unit to run at high CO₂ excess and achieve a more power-efficient CO conversion. This is enabled by the membrane separation's comparatively sharp cut and low cost. The pressure-swing adsorption's poor CO recovery of 86% dilutes the electrolysis feed stream, lowering the CO₂ partial pressure and pushing the electrolyzer to a less efficient operating point. Part of the issue originates from the modeling approach: The pressure-swing adsorption is modeled as a simplified two-step cycle, which lacks a rinse/blowdown phase, ordinarily used to increase the CO recovery. Finally, the amine washing unit is simply too expensive, its large reboiler duty preventing the CO₂ electrolyzer from running at high CO₂ excess.

Additionally, the model cannot deliver satisfactory syngas purity (94%) and leaves a significant amount of CO₂ in the product stream. Across all three flowsheets, roughly half of the reacted CO₂ is lost homogeneously to the anode side, creating a drag on syngas production costs. Membrane separation is therefore the most economical downstream option for electrolytic syngas production, primarily because it best enables an efficient CO₂ electrolyzer.

A sensitivity analysis was conducted in which electricity price, CO₂ electrolyzer cost, and CO₂ feed price were varied. Electricity price is the biggest contributor to syngas production costs, the CO₂ electrolyzer cost has a sizeable influence as well, and the CO₂ feed price's influence is minor even at 0 €/t_{CO₂}. The separation unit ranking stayed the same in every scenario.

In future improvements the pressure-swing adsorption model needs to be extended to include a blowdown phase. The additional CO recovery from the blowdown phase could make the pressure-swing adsorption competitive with the membrane flowsheet. The CO₂ electrolyzer model inherited the M3 model's limits (atmospheric pressure, no H₂ inlet, fixed configuration); a closer study into which electrolyzer type to use, and a corresponding model, would give deeper insight into the electrolyzer–separation coupling. To improve comparability between the separation alternatives, an amine wash model that reaches the required 98% purity could be implemented. Finally, CO₂ electrolyzer types with a liquid inlet could also be considered, as these configurations require less separation effort.

List of Abbreviations and Symbols

Abbreviations

AEM	Anion Exchange Membrane
BPM	Bipolar Membrane
BVP	Boundary Value Problem
CAPEX	Capital Expenditure
CEM	Cation (Proton) Exchange Membrane
CEPCI	Chemical Engineering Plant Cost Index
CL	Catalyst Layer
CSS	Cyclical Steady State
DAE	Differential-Algebraic Equation
DEA	Diethanolamine
DM	Diffusion Medium
FE	Faradaic Efficiency
GDE	Gas Diffusion Electrode
HER	Hydrogen Evolution Reaction
HHV	Higher Heating Value
IPOPT	Interior Point Optimizer
LHS	Latin Hypercube Sampling

M1, M2, M3	Electrolysis models of increasing depth (Bagemihl et al.)
MDEA	N-Methyldiethanolamine
MEA	Monoethanolamine
M&S	Marshall & Swift Equipment Cost Index
NIST	National Institute of Standards and Technology
NLS	Nonlinear Least Squares
OER	Oxygen Evolution Reaction
OPEX	Operational Expenditure
PEM	Polymer Electrolyte (Proton Exchange) Membrane
PSA	Pressure Swing Adsorption
PTL	Porous Transport Layer
PVF	Present Value Factor
PySR	Python Symbolic Regression package
PZ	Piperazine
TEA	Techno-Economic Assessment
TEC	Total Equipment Cost
VMB	Virtual Moving Bed
VSA	Vacuum Swing Adsorption

Latin Symbols

A	Electrode / heat-transfer area	$[\text{m}^2]$
A_{tot}	Total electrode area	$[\text{m}^2]$
A_{mem}	Membrane area	$[\text{m}^2]$
$A_i \dots F_i$	Shomate equation coefficients	[various]
b_i	Sips isotherm affinity constant	$[\text{bar}^{-1}]$

BC_{membrane}	Area-specific membrane base cost	[€/m ²]
$BV_{i,CO}$	Butler–Volmer exchange current density, CO formation	[A/m ²]
BV_{i,H_2}	Butler–Volmer exchange current density, HER	[A/m ²]
BV_{CO}	Butler–Volmer transfer coefficient, CO formation	[–]
BV_{H_2}	Butler–Volmer transfer coefficient, HER	[–]
BV_{ECO}	Cathode equilibrium potential, CO formation	[V]
$BV_{CO_2,ref}$	Reference CO ₂ concentration (kinetics)	[mol/m ³]
c_i	Molar concentration of species i	[mol/m ³]
C_{CAPEX}	Total capital expenditure	[€]
$C_{\text{CAPEX},x}$	Capital expenditure of component x	[€]
c_{area}	Electrolyzer area cost	[€/m ²]
$\bar{c}_{CO_2}$	Mean cathode CO ₂ concentration	[mol/m ³]
$c_{\text{el}}, p_{\text{el}}$	Electricity price	[€/kWh]
C_{kW}	Specific capital cost of H ₂ electrolyzer	[€/kW]
$C_{\text{lifecycle}}$	Total lifecycle cost	[€]
$\dot{C}_{\text{OPEX}}$	Total annual operational expenditure	[€/a]
$\dot{C}_{CO_2,feed}$	Annual fresh CO ₂ feedstock cost	[€/a]
$\dot{C}_{\text{OPEX},x}$	Annual operational expenditure of component x	[€/a]
C_p	Purchased equipment cost	[\$]
$c_{p,mix}$	Molar heat capacity of the mixture	[J/(mol K)]
c_{steam}	Steam cost	[€/GJ]
D	Vessel diameter / diffusion coefficient	[ft, m ² /s]
E_{anode}	Anode equilibrium potential	[V]
E_{appl}	Applied cathode voltage	[V]

e_{H_2}	Specific energy consumption of H_2 electrolyzer	[kWh/kg]
F	Faraday constant	[C/mol]
F_{BM}	Bare module factor	[—]
FE_i	Faradaic efficiency of species i	[—]
F	Volumetric feed-gas flow rate (amine model)	[$10^3 \text{ Nm}^3/\text{h}$]
f_{maint}	Maintenance factor	[—]
f_{stack}	Stack replacement factor	[—]
H	Channel / vessel height	[mm, ft]
$ha/\rho_s C_s$	Overall heat-transfer coefficient (kinetics)	[s^{-1}]
H_c, H_{ch}	Catalyst layer / electrode thickness	[mm]
HHV_{H_2}	Higher heating value of hydrogen	[kWh/kg]
h_{op}	Annual operating hours	[h/a]
I	Total current	[A]
i	Current density	[A/m ²]
J_i	Membrane molar flux of species i	[mol/(m ² s)]
K_0, K_1, K_2, K_3	Cost-correlation coefficients	[—]
K_G	Guthrie base vessel cost	[$10^3 \text{ \$}$]
k_H	Henry constant	[—]
L	Channel / vessel length	[mm, ft]
L_m	Membrane thickness	[m]
$\dot{m}_{\text{CO}_2, \text{cap}}$	Captured CO_2 mass rate (amine model)	[t/h]
$\dot{m}_{H_2}$	Hydrogen mass flow	[kg/s]
M_t, M_∞	Adsorbed loading at time t / at equilibrium	[—]
N	Number of channels	[—]

n	Plant lifetime / Sips heterogeneity exponent	[a, —]
$\dot{n}$	Molar flow rate	[mol/s]
N_{trains}	Number of (relaxed) PSA columns	[—]
P, p	Pressure	[bar, kPa]
P_{el}	Power consumption	[W]
p_H, p_L	Adsorption / desorption pressure	[kPa]
P_{hp}	H ₂ electrolyzer power draw	[kW]
P_i	Membrane permeability of species i	[barrer]
p_r, p_p	Retentate / permeate pressure	[bar]
Q	Total charge passed	[C]
q_i^*	Equilibrium loading of species i	[mol/kg]
$\dot{Q}$	Thermal duty	[W]
Q_i	Membrane permeance of species i	[mol/(m ² s Pa)]
$q_{m,i}$	Sips saturation capacity	[mol/kg]
R	Universal gas constant	[J/(mol K)]
r	Discount / interest rate	[—]
R^2	Coefficient of determination	[—]
T	Temperature	[K]
TEC	Total equipment cost (amine model)	[M€]
U	Overall heat-transfer coefficient	[W/(m ² K)]
u_l	Electrolyte flow speed	[ml/min]
v_{gas}	Gas flow speed	[ml/min]
V_{cell}	Cell voltage	[V]
$\dot{V}$	Volumetric gas flow rate	[m ³ /s]

W	Channel width	[mm]
$\dot{W}$	Shaft power	[kW]
$\mathbf{x}_{\text{opt}}$	Vector of optimization design variables	[—]
x_i, y_i, z_i	Mole fraction of species i	[—]

Greek Symbols

α, β	Guthrie cost-correlation exponents	[—]
$\alpha_{i,j}$	Amine duty surrogate coefficients	[—]
$\alpha_{\text{TEC}}, \beta_{\text{TEC}}, \gamma_{\text{TEC}}$	Amine TEC regression coefficients	[—]
β'	Kinetic correction factor	[—]
$\chi_{\text{tot}}, \chi_{\text{het}}, \chi_{\text{hom}}$	Total / heterogeneous / homogeneous conversion	[—]
δ	Membrane selective-layer thickness	[μm]
η	Isentropic / pump / compressor efficiency	[—]
$\eta_{\text{act,A}}$	Anodic activation overpotential	[V]
η_{el}	Overall electrical efficiency	[—]
η_{ohm}	Ohmic overpotential	[V]
γ	Ratio of specific heats c_p/c_v	[—]
$\kappa_0 \dots \kappa_3$	H ₂ electrolyzer cost-correlation coefficients	[—]
λ_{CO_2}	CO ₂ stoichiometric ratio (excess)	[—]
$\omega_1 \dots \omega_8$	Compressor cost-correlation constants	[—]
ρ_s	Adsorbent solid density	[kg/m ³]
$\sigma_{\text{el}}, \sigma_m$	Electrode / membrane conductivity	[S/m]
$\theta_{i,j}$	Symbolic-regression surrogate constants	[—]

Subscripts and Indices

act	Activation
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amine	Amine washing unit
appl	Applied
cap	Captured
cell	Per single channel / cell
CO ₂	Carbon dioxide
CO	Carbon monoxide
comp	Compressor
des	Desorption
el, elec	Electrolyzer / electricity
feed	Fresh CO ₂ feed
gas, in	Gas inlet
H ₂	Hydrogen
HX	Heat exchanger
in / out	Inlet / outlet
maint	Maintenance
mem	Membrane
ohm	Ohmic
opt	Optimization variable
p, r	Permeate / retentate
pump	Vacuum pump
reb	Reboiler
sep	Separation unit
tot	Total

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Declaration of AI Tools

The following AI tools were used in this thesis.

Section	Tool(s) used	Reason for use
All chapters	Overleaf AI, Claude	Grammar / spell checking
All chapters	Claude	Restructuring tables, figures, equations in LaTeX
Chapters 3, 4, 5	Gemini, Claude Code	Generating code to plot results
Chapter 3	Claude Code	Assisting in wiring flowsheets together
Conclusion, Intro	Claude	Reformulating / restructuring text
Conclusion, Intro	Claude	Cross-checking against provided writing tips

A Appendix

A.1 Surrogate Model Coefficients

Table A.1: Coefficients of the CO₂ electrolysis surrogate equations

Equation	Symbol	Value
$c_{CO_2,out}$	$\theta_{1,1}$	-3.7203176
	$\theta_{1,2}$	1.3941393
	$\theta_{1,3}$	40.244705
$c_{CO,out}$	$\theta_{2,1}$	1.4358697
	$\theta_{2,2}$	-0.46066585
	$\theta_{2,3}$	-0.6278232
	$\theta_{2,4}$	21.189863
	$\theta_{2,5}$	-1.1215197
	$\theta_{2,6}$	10.42661
$c_{H_2,out}$	$\theta_{3,1}$	0.26091927
	$\theta_{3,2}$	0.35308436
	$\theta_{3,3}$	14.994374

A.2 Butler–Volmer Kinetic Parameters

Table A.2: Butler–Volmer kinetic parameters of the CO₂ electrolysis voltage–current model (equation (3.10)–equation (3.16)), adopted from Bagemihl et al. [10]

Symbol	Description	Value
$BV_{i,CO}$	Exchange current density, CO formation (A/m ²)	3.3×10^{-4}
BV_{i,H_2}	Exchange current density, HER (A/m ²)	3.4×10^{-6}
BV_{CO}	Transfer coefficient, CO formation (–)	0.33
BV_{H_2}	Transfer coefficient, HER (–)	0.33
BV_{ECO}	Cathode equilibrium potential, CO formation (V)	0.11
$BV_{CO_2,ref}$	Reference CO ₂ concentration (mol/m ³)	34

A.3 H₂ Electrolyzer Specific Capital Cost Coefficients

Table A.3: Fitted coefficients of the H₂ electrolyzer specific capital cost correlation [28] (equation (3.25))

Symbol	Description	Value
κ_3	Cubic coefficient	–38.131
κ_2	Quadratic coefficient	433.69
κ_1	Linear coefficient	–1690
κ_0	Constant term	3634.6

A.4 CEPCI Values for Cost Index Scaling

Table A.4: CEPCI values used for cost index scaling

Year	CEPCI
1969	119.0
1974	144.1
1993	359.2
2001	397.0
2021	708.8
2024	797.9

A.5 Compressor Cost Correlation Constants

Table A.5: Constants for the Guthrie compressor cost correlation [30]

Symbol	Description	Value
ω_1	Base cost coefficient (USD, 1974)	23,000
ω_2	Reference shaft power (W)	73,550
ω_3	Base cost exponent	0.77
ω_4	Accessories cost coefficient	776.5
ω_5	Accessories cost exponent	0.493
ω_6	Installation factor $f_{install}$	1.565
ω_7	Indirect cost factor f_{ind}	1.38
ω_8	USD to EUR conversion $r_{USD/EUR}$	0.96

A.6 Amine Washing Surrogate Coefficients

Table A.6: Coefficients of the total equipment cost regression [39] (equation (3.59))

Symbol	Description	Value
α_{TEC}	Constant term	2.1673
β_{TEC}	Composition coefficient	0.8092
γ_{TEC}	Flow coefficient	−0.00332
n_{TEC}	Composition exponent	0.5291
m_{TEC}	Flow exponent	0.8391

Table A.7: Coefficients of the amine washing specific-duty surrogates [39] (equation (3.64)). The reboiler and cooling duties are specific to the captured CO_2 in $\text{GJ/t}_{\text{CO}_2}$, the electrical duty in $\text{kWh/t}_{\text{CO}_2}$; $\alpha_{i,1}$ and $\alpha_{i,3}$ carry these units, $\alpha_{i,2}$ and $\alpha_{i,4}$ are dimensionless.

Duty	$\alpha_{i,1}$	$\alpha_{i,2}$	$\alpha_{i,3}$	$\alpha_{i,4}$
$\dot{Q}_{\text{reb}} (i = 1)$	1.471	−35.83	3.560	−0.0158
$\dot{W}_{\text{elec}} (i = 2)$	10.420	−23.49	2.164	−1.2350
$\dot{Q}_{\text{cool}} (i = 3)$	10.040	−33.73	2.905	0.2108

A.7 Additional CO_2 Electrolysis Sensitivity Plots

The following figures complete the sensitivity analysis of chapter 4, showing the remaining input-output combinations around the operating point $E_{\text{appl}} = -1.25 \text{ V}$, $v_{\text{gas}} = 27.5 \text{ ml/min}$, $\text{CO}_{2,\text{share}} = 0.8$.

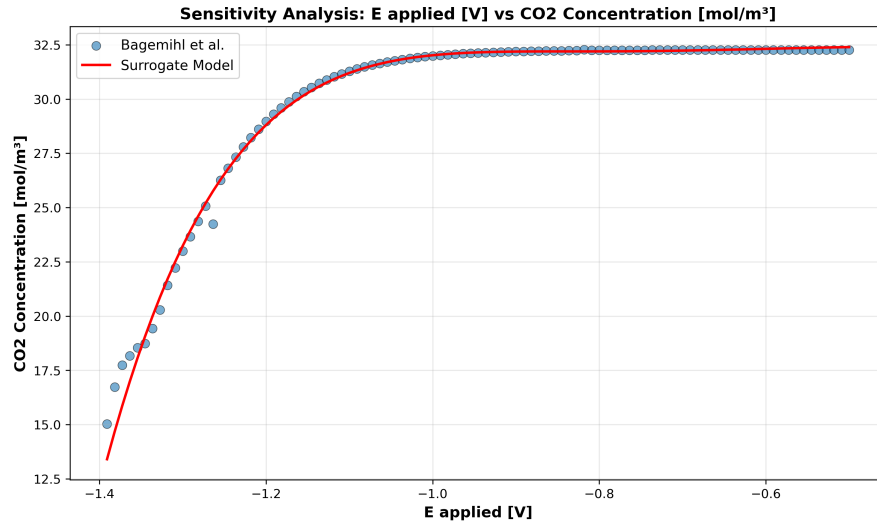


Figure A.1: Sensitivity analysis: Effect of applied voltage E_{appl} on CO₂ concentration output at $CO_{2\text{share}} = 0.8$ and $v_{\text{gas}} = 27.5$ ml/min

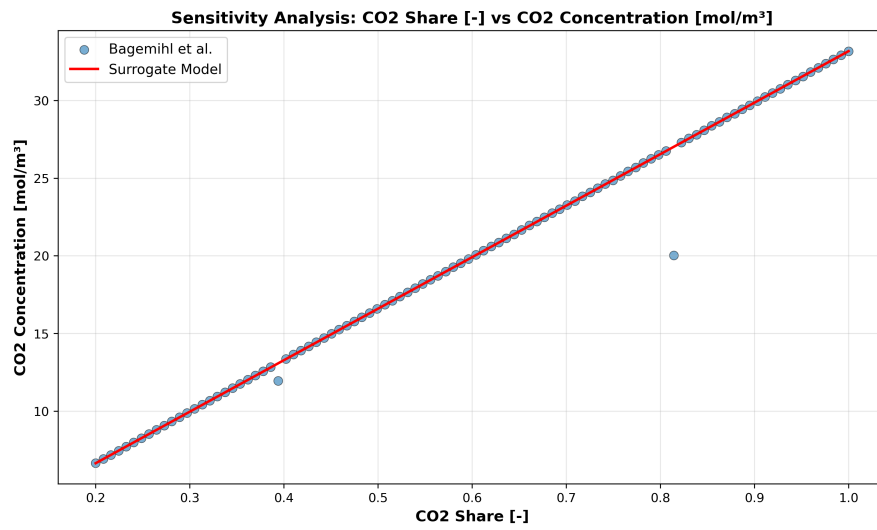


Figure A.2: Sensitivity analysis: Effect of CO₂ share on CO₂ concentration output at $E_{\text{appl}} = -1.25$ V and $v_{\text{gas}} = 27.5$ ml/min

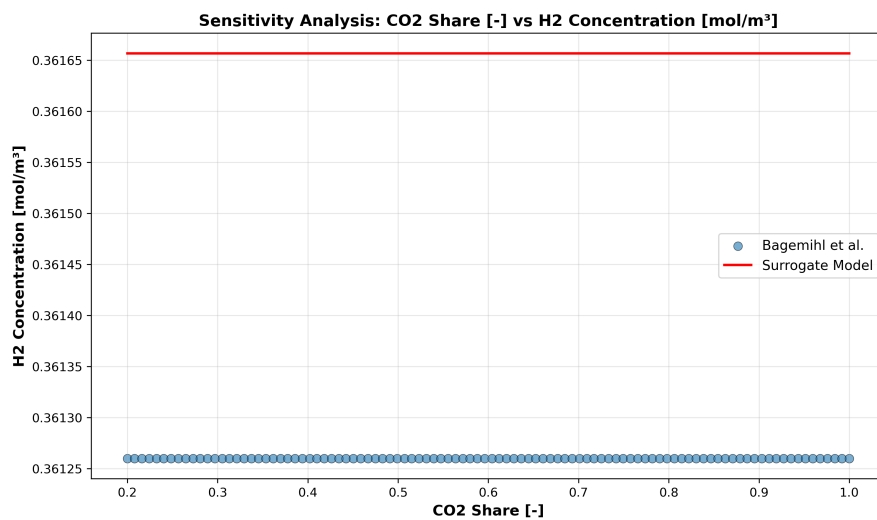


Figure A.3: Sensitivity analysis: Effect of CO₂ share on H₂ concentration output at $E_{\text{appl}} = -1.25 \text{ V}$ and $v_{\text{gas}} = 27.5 \text{ ml/min}$

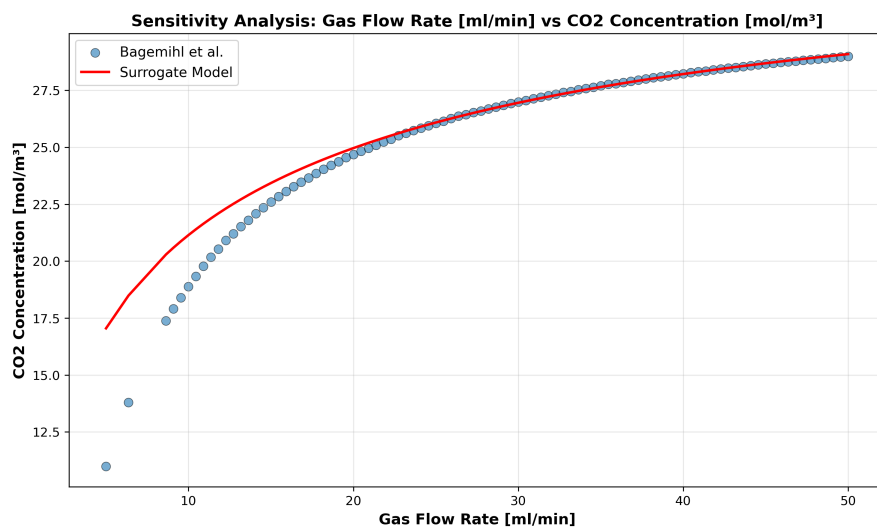


Figure A.4: Sensitivity analysis: Effect of gas flow rate v_{gas} on CO₂ concentration output at $E_{\text{appl}} = -1.25 \text{ V}$ and $\text{CO}_{2\text{share}} = 0.8$

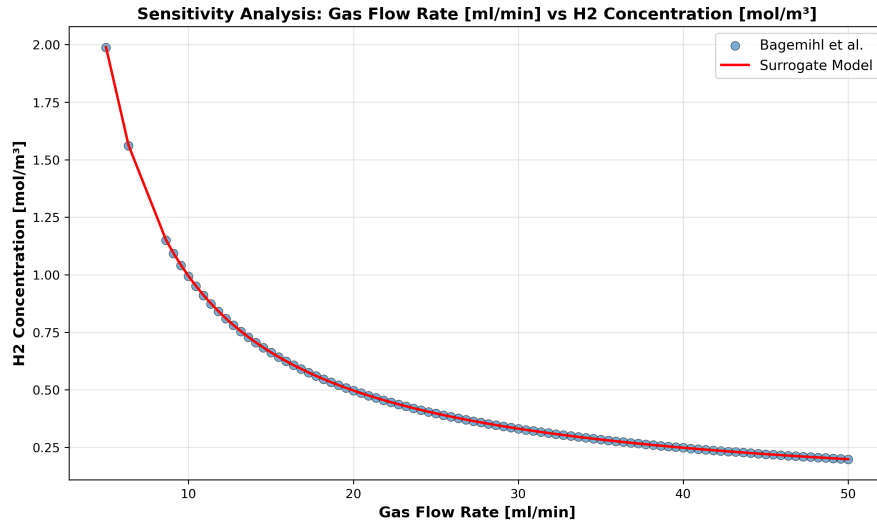


Figure A.5: Sensitivity analysis: Effect of gas flow rate v_{gas} on H_2 concentration output at $E_{appl} = -1.25$ V and $CO_{2share} = 0.8$

A.8 Additional PSA Parametric Sweep Plots

The following figures show further internal operating variables of the PSA model obtained from the parametric sweep described in section 4.2, plotted against the adsorption pressure.

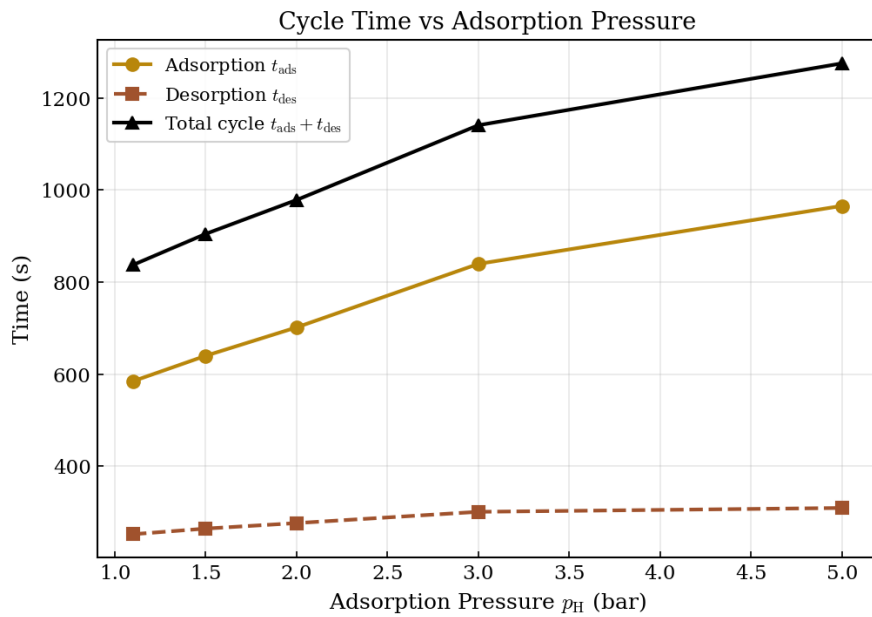


Figure A.6: PSA: Cycle Time vs Adsorption Pressure

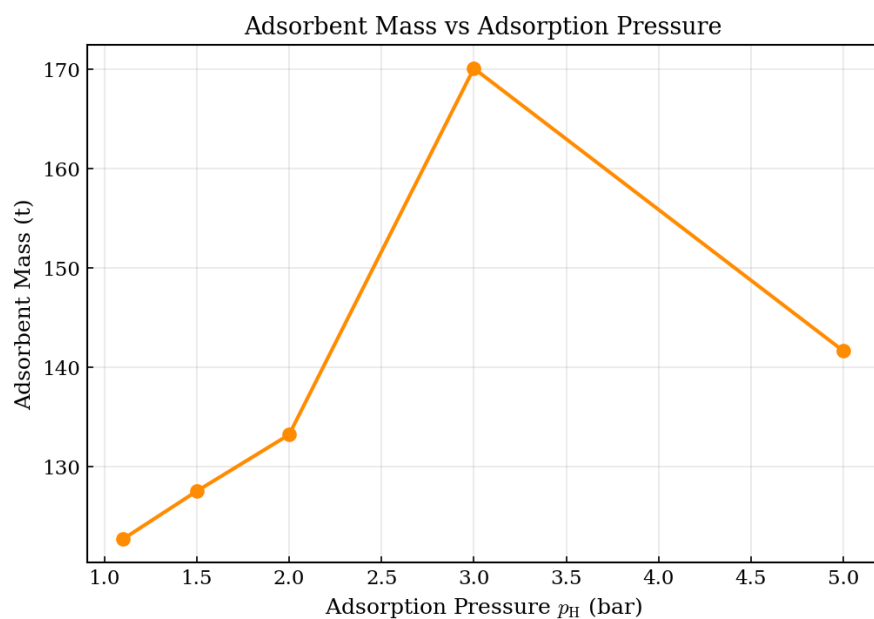


Figure A.7: PSA: Adsorbent Mass vs Adsorption Pressure

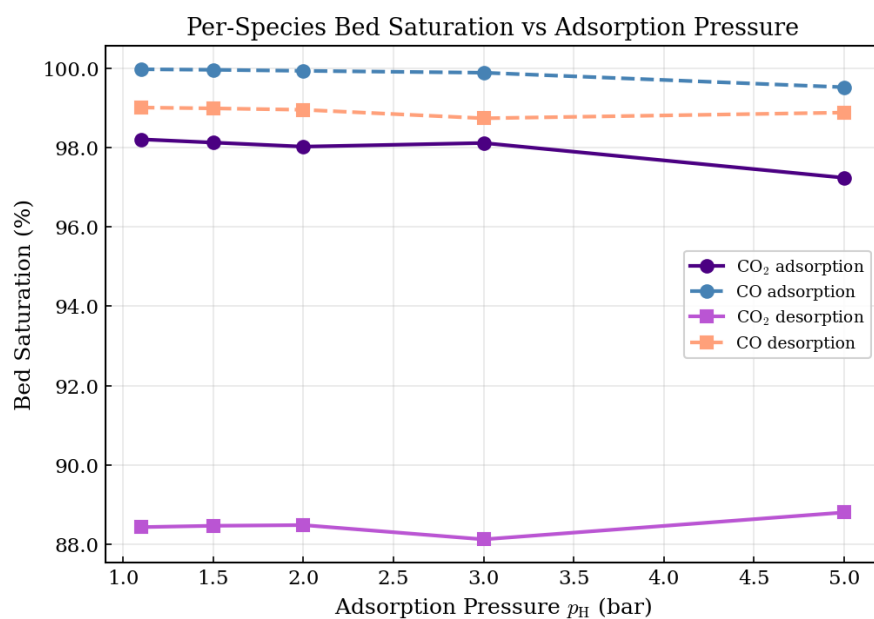


Figure A.8: PSA: Bed Saturation vs Adsorption Pressure

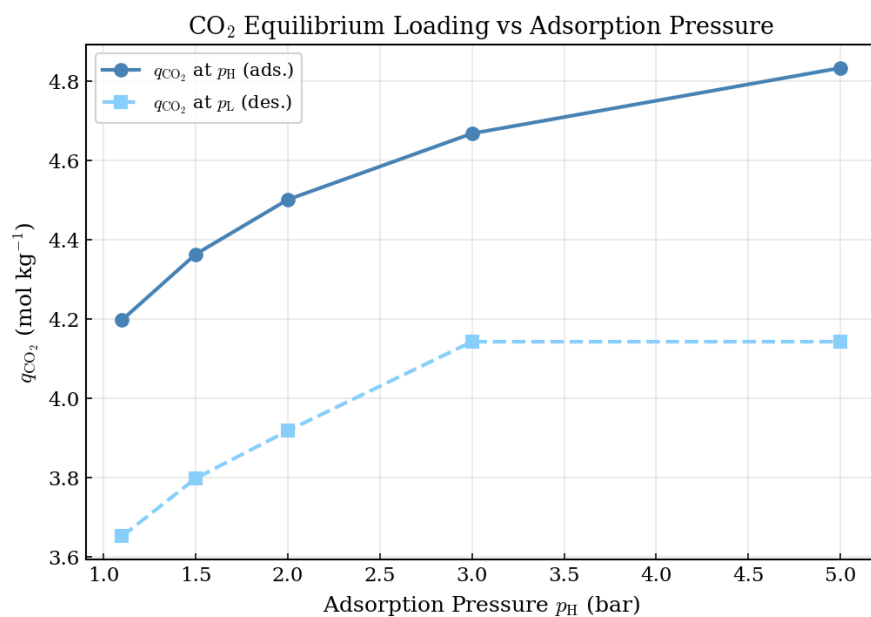
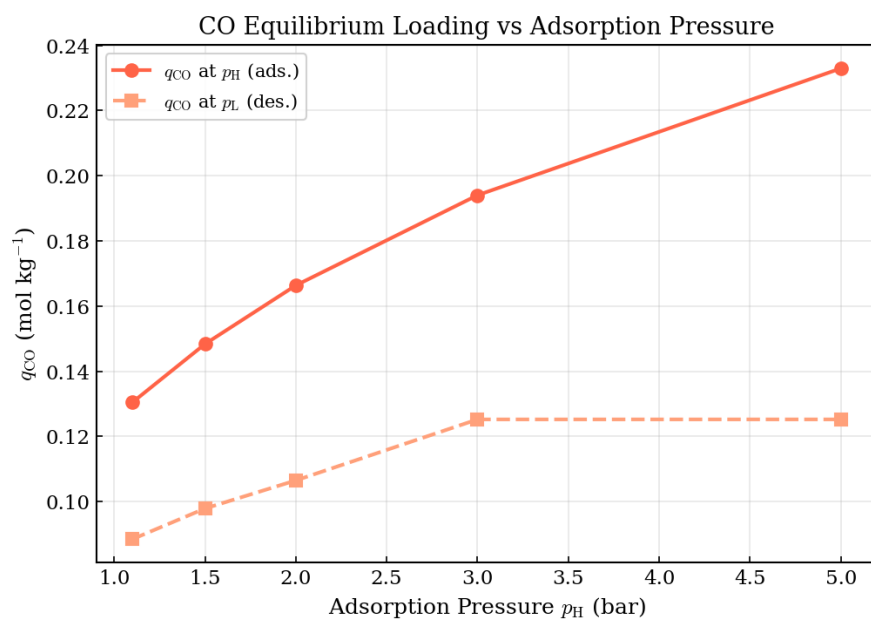
Figure A.9: PSA: CO₂ Equilibrium Loading vs Adsorption Pressure

Figure A.10: PSA: CO Equilibrium Loading vs Adsorption Pressure

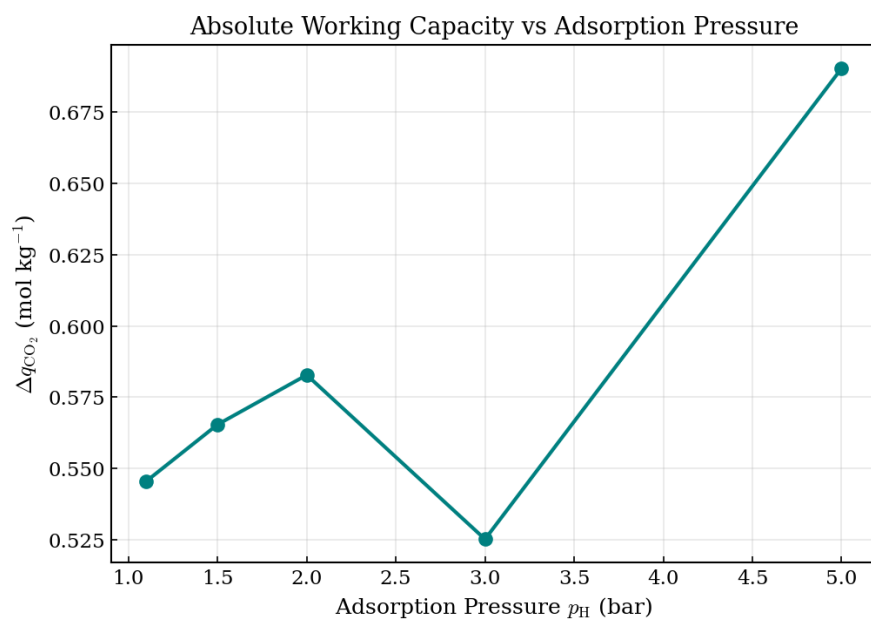


Figure A.11: PSA: Absolute Working Capacity vs Adsorption Pressure

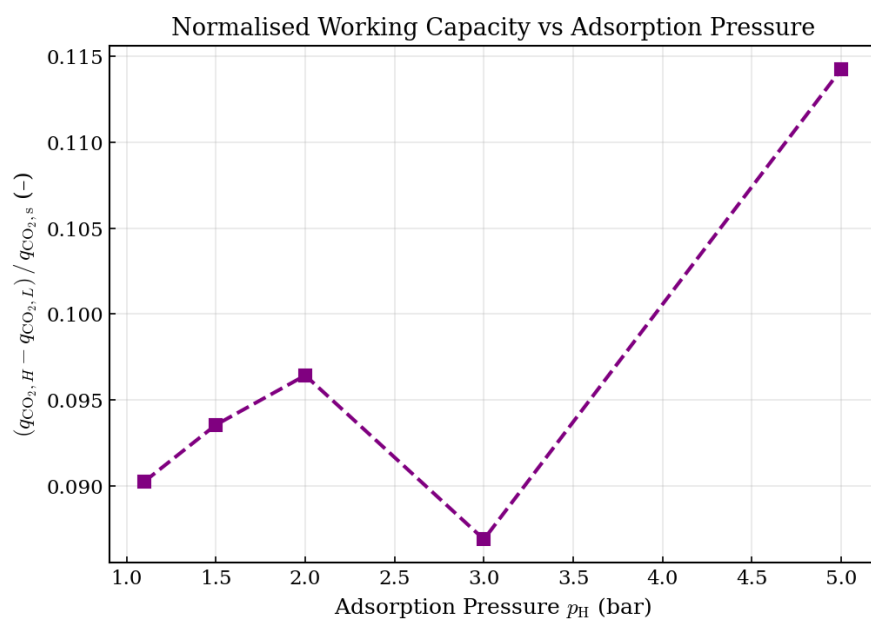


Figure A.12: PSA: Normalized Working Capacity vs Adsorption Pressure